

Lecture 16: Linear Dimensionality Reduction for Element Properties, Spectra and Images

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Principal Component Analysis

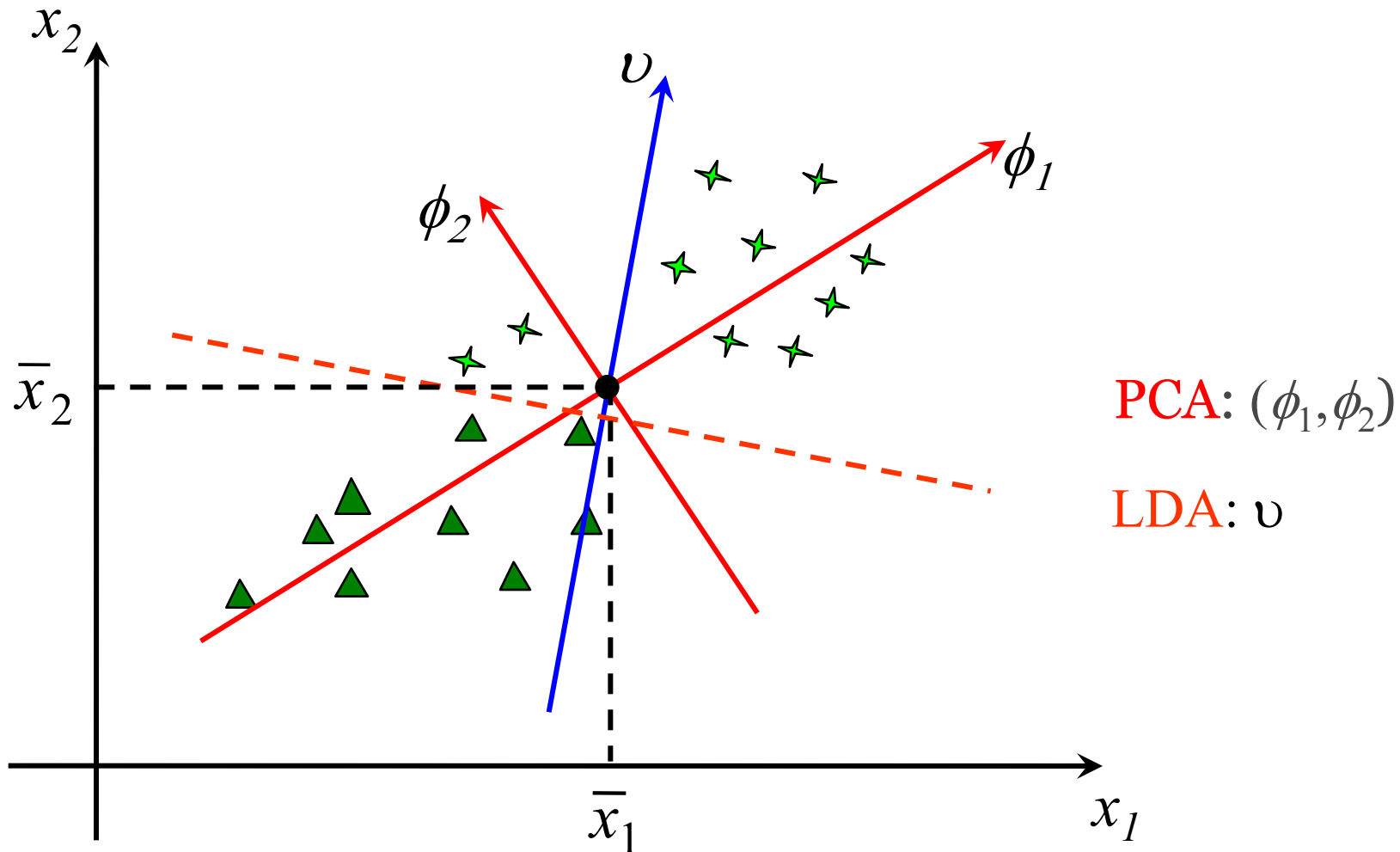
$$S(\mathbf{x}, \mathbf{R}) = \sum_i a_i(\mathbf{x}) w_i(\mathbf{R}) + N$$

- The first PC retains the greatest amount of variation in the sample
- The k^{th} PC retains the k^{th} greatest fraction of the variation in the sample
- The k^{th} largest eigenvalue of the correlation matrix C is the variance in the sample along the k^{th} PC

Linear Discriminant Analysis

- Linear Discriminant Analysis, or simply LDA, is a feature extraction technique that has been used successfully in many statistical pattern recognition problems.
- LDA is often called Fisher Discriminant Analysis (FDA).
- The primary purpose of LDA is to separate samples of distinct groups by transforming them to a space which maximises their between-class separability while minimising their within-class variability.
- It assumes implicitly that the true covariance matrices of each class are equal because the same within-class scatter matrix is used for all the classes considered.
- If we remove this assumption, we get Quadratic Discriminant Analysis

Geometric Idea of LDA



LDA Steps

1. Compute the d -dimensional mean vectors.
2. Compute the scatter matrices
3. Compute the eigenvectors and corresponding eigenvalues for the scatter matrices.
4. Sort the eigenvalues and choose those with the largest eigenvalues to form a $d \times k$ dimensional matrix
5. Transform the samples onto the new subspace.

LDA Method

- Let the between-class scatter matrix S_b be defined as

$$S_b = \sum_{i=1}^g N_i (\bar{x}_i - \bar{x})(\bar{x}_i - \bar{x})^T$$

- and the within-class scatter matrix S_w be defined as

$$S_w = \sum_{i=1}^g (N_i - 1) S_i = \sum_{i=1}^g \sum_{j=1}^{N_i} (x_{i,j} - \bar{x}_i)(x_{i,j} - \bar{x}_i)^T$$

- where $x_{i,j}$ is an n -dimensional data point j from class p_i , N_i is the number of training examples from class p_i , and g is the total number of classes or groups.

LDA Method

- The sample mean, sample covariance, and grand mean vector are given respectively by:

$$\bar{x}_i = \frac{1}{N_i} \sum_{j=1}^{N_i} x_{i,j}$$

$$S_i = \frac{1}{(N_i - 1)} \sum_{j=1}^{N_i} (x_{i,j} - \bar{x}_i)(x_{i,j} - \bar{x}_i)^T$$

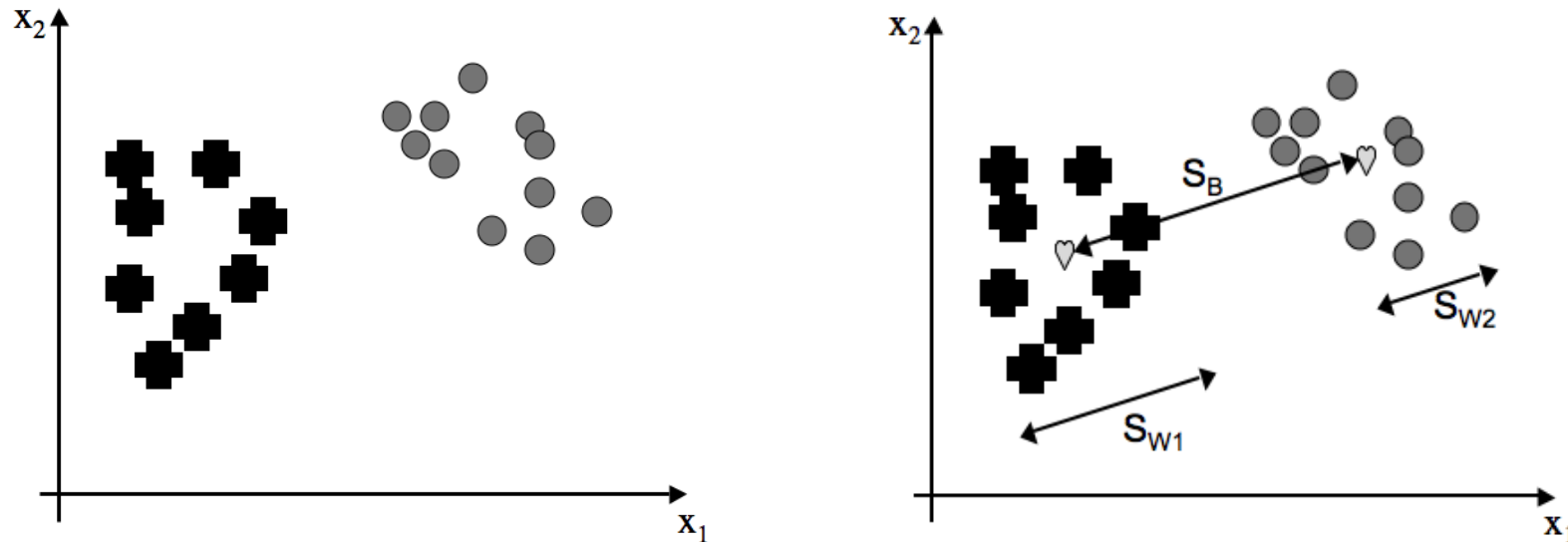
$$\bar{x} = \frac{1}{N} \sum_{i=1}^g N_i \bar{x}_i = \frac{1}{N} \sum_{i=1}^g \sum_{j=1}^{N_i} x_{i,j}$$

$$N = N_1 + N_2 + \dots + N_g$$

LDA Method

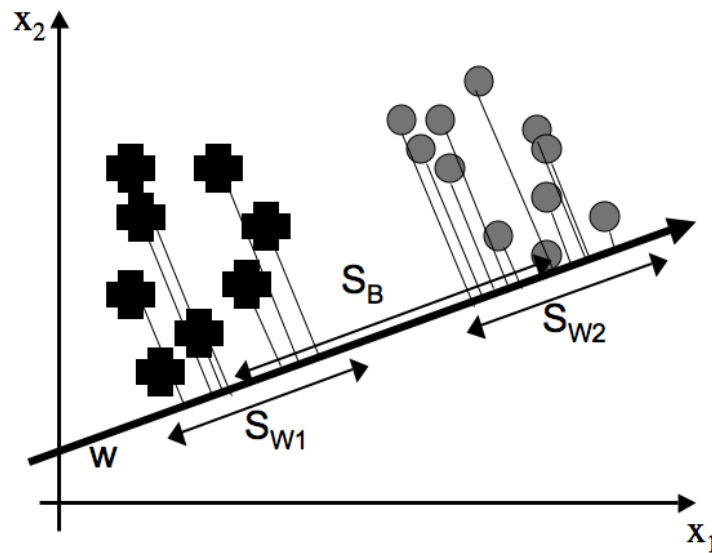
The objective of LDA is to find a projection matrix P_{lda} that maximises the ratio of the determinant of S_b to the determinant of S_w (Fisher's criterion) as:

$$P_{lda} = \arg \max_P \frac{|P^T S_b P|}{|P^T S_w P|}$$

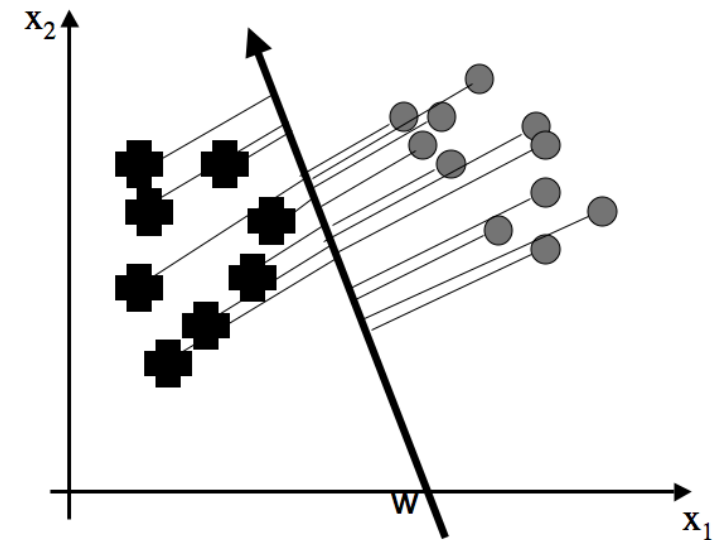


LDA Method

- So Fisher's criterion tries to find the projection that:
 - Maximises the variance of the class means
 - Minimises the variance of the individual classes



Good projection



Bad projection

Classification using LDA

- The LDA is an axis projection.
- Once the projection is found all the data points can be transformed to the new axis system along with the class means and covariances.
- Allocation of a new point to a class can be done using a distance measure such as the Mahalanobis distance.

Given a vector $\mathbf{x}$, a mean vector μ , and a covariance matrix Σ , the Mahalanobis Distance D_M is defined as:

$$D_M(\mathbf{x}, \mu) = \sqrt{(\mathbf{x} - \mu)^T \Sigma^{-1} (\mathbf{x} - \mu)}$$

where:

- $\mathbf{x}$ is the vector for which the Mahalanobis Distance is being computed.
- μ is the mean vector of the distribution.
- Σ is the covariance matrix of the distribution.
- T denotes a matrix transpose.
- Σ^{-1} is the inverse of the covariance matrix.

LDA vs. PCA

- LDA is supervised ML method, PCA is unsupervised method
- LDA seeks directions that are efficient for *discriminating* data whereas PCA seeks directions that are efficient for *representing* data.
- The directions that are discarded by PCA might be exactly the directions that are necessary for distinguishing between groups.

`sklearn.discriminant_analysis.LinearDiscriminantAnalysis`

```
class sklearn.discriminant_analysis.LinearDiscriminantAnalysis(solver='svd', shrinkage=None, priors=None,
n_components=None, store_covariance=False, tol=0.0001, covariance_estimator=None)
```

[\[source\]](#)

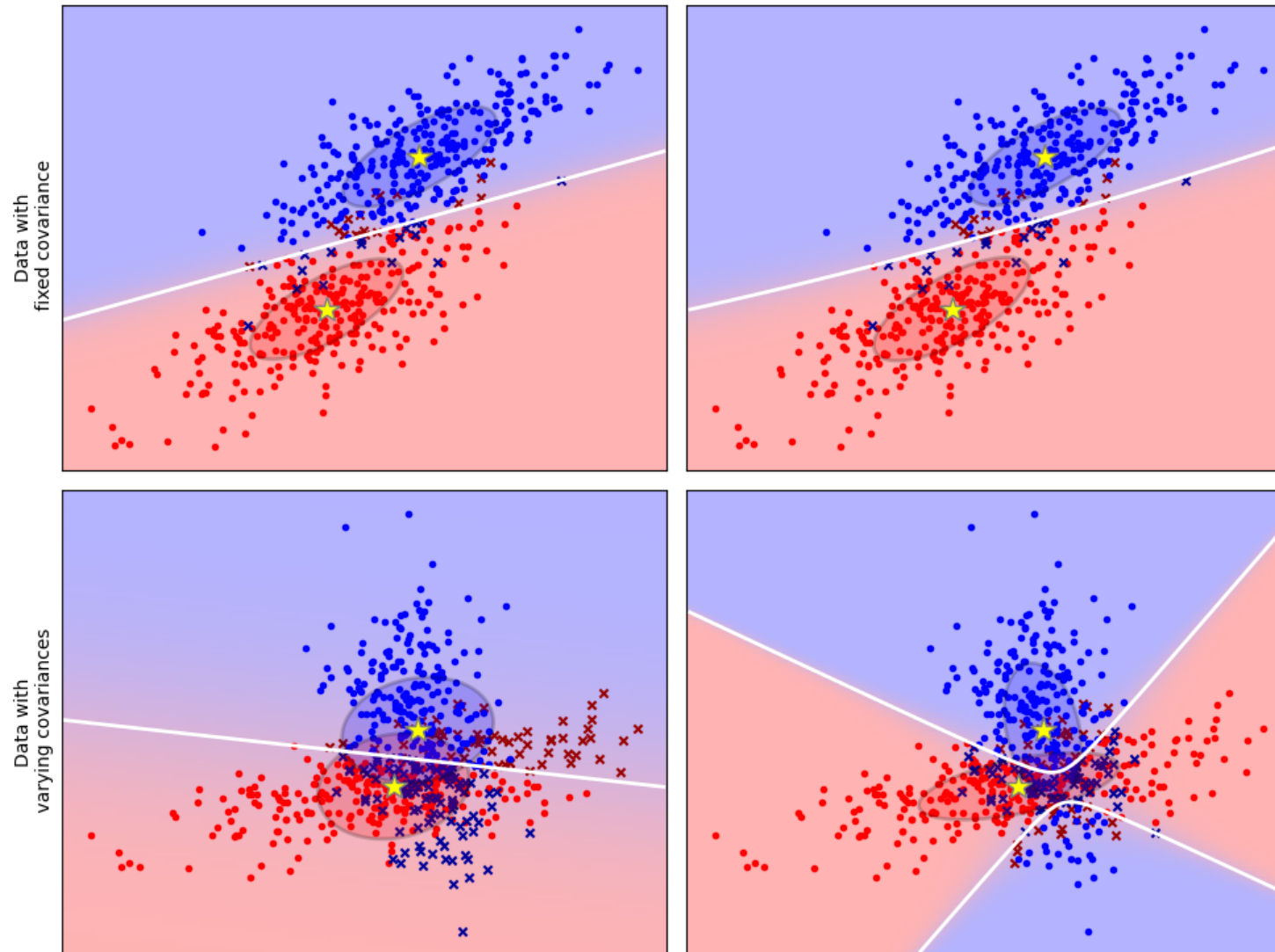
<code>decision_function(X)</code>	Apply decision function to an array of samples.
<code>fit(X, y)</code>	Fit the Linear Discriminant Analysis model.
<code>fit_transform(X[, y])</code>	Fit to data, then transform it.
<code>get_feature_names_out([input_features])</code>	Get output feature names for transformation.
<code>get_metadata_routing()</code>	Get metadata routing of this object.
<code>get_params([deep])</code>	Get parameters for this estimator.
<code>predict(X)</code>	Predict class labels for samples in X.
<code>predict_log_proba(X)</code>	Estimate log probability.
<code>predict_proba(X)</code>	Estimate probability.
<code>score(X, y[, sample_weight])</code>	Return the mean accuracy on the given test data and labels.
<code>set_output(*[, transform])</code>	Set output container.
<code>set_params(**params)</code>	Set the parameters of this estimator.
<code>set_score_request(*[, sample_weight])</code>	Request metadata passed to the <code>score</code> method.
<code>transform(X)</code>	Project data to maximize class separation.

Quadratic Discriminant Analysis

Linear Discriminant Analysis vs Quadratic Discriminant Analysis

Linear Discriminant Analysis

Quadratic Discriminant Analysis



LDA vs. QDA

Bayesian classification framework:

$$P(y = k|x) = \frac{P(x|y = k)P(y = k)}{P(x)} = \frac{P(x|y = k)P(y = k)}{\sum_l P(x|y = l) \cdot P(y = l)}$$

QDA approximation:

$$P(x|y = k) = \frac{1}{(2\pi)^{d/2}|\Sigma_k|^{1/2}} \exp\left(-\frac{1}{2}(x - \mu_k)^t \Sigma_k^{-1}(x - \mu_k)\right)$$

LDA is a special case of QDA, where the Gaussians for each class are assumed to share the same covariance matrix: $\Sigma_k = \Sigma$. If so

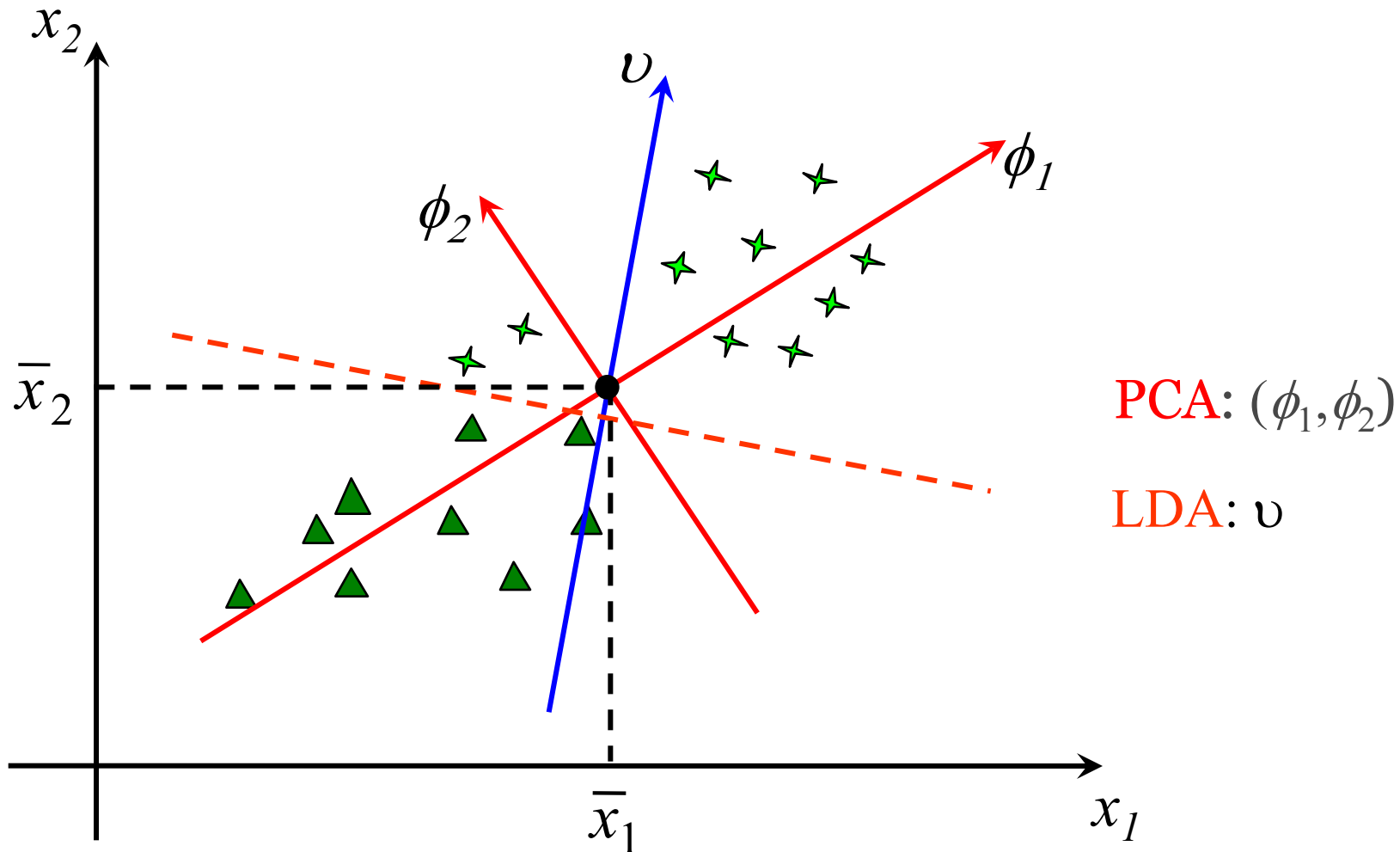
$$\log P(y = k|x) = \omega_k^t x + \omega_{k0} + Cst.$$

where $\omega_k = \Sigma^{-1} \mu_k$ and $\omega_{k0} = -\frac{1}{2} \mu_k^t \Sigma^{-1} \mu_k + \log P(y = k)$. These quantities correspond to the `coef_` and `intercept_` attributes, respectively.

Note: Relation with Gaussian Naive Bayes

If in the QDA model one assumes that the covariance matrices are diagonal, then the inputs are assumed to be conditionally independent in each class, and the resulting classifier is equivalent to the Gaussian Naive Bayes classifier `naive_bayes.GaussianNB`.

Geometric Idea of PCA and LDA



Spectroscopic Imaging

Scanning probe microscopy:

- Force-distance curve measurements
- Current-voltage measurements
- Piezoresponse force/electrochemical strain spectroscopy

Electron microscopy:

- Electron Energy Loss Spectroscopy

Optical microscopy:

- Hyperspectral imaging
- Time resolved measurements

Mass-spectrometry:

- Secondary ion MS imaging

In many cases, measured signal can be represented or approximated as a linear combination of signals. However, their functional forms are generally unknown

Very important: convolution with resolution function is also mixing

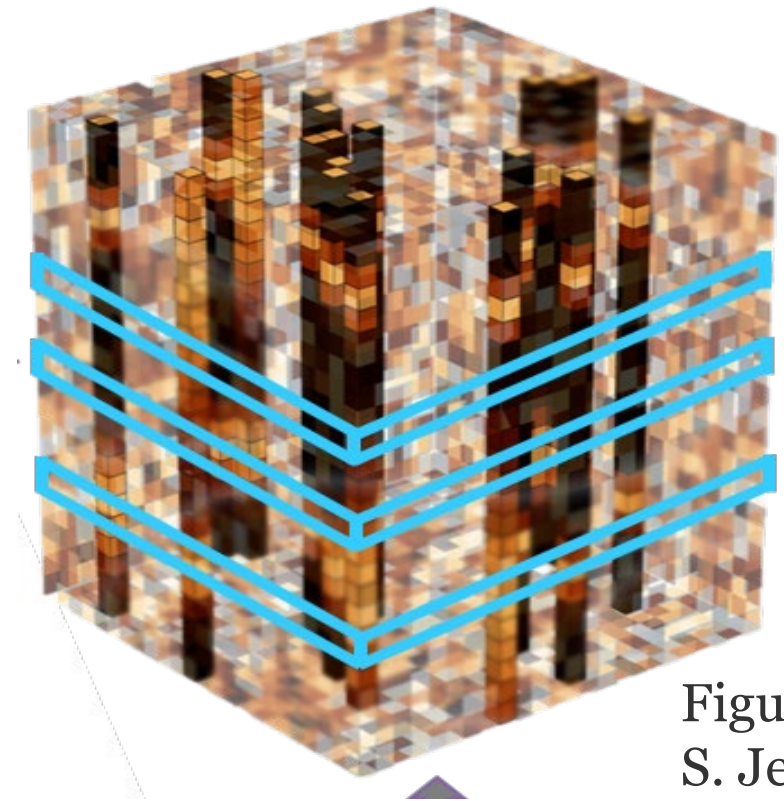
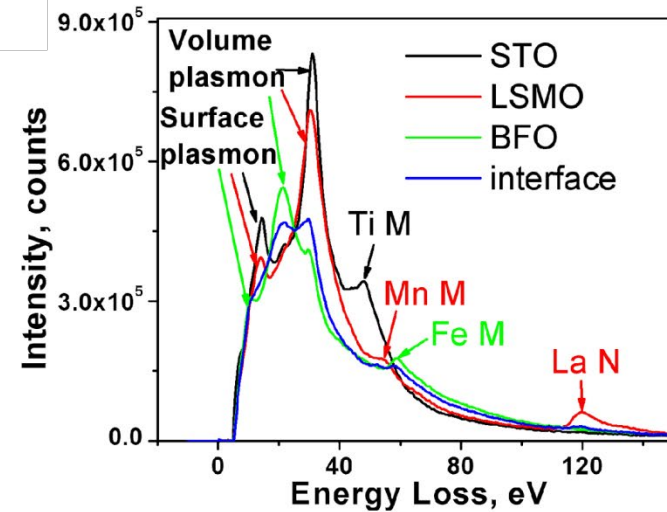
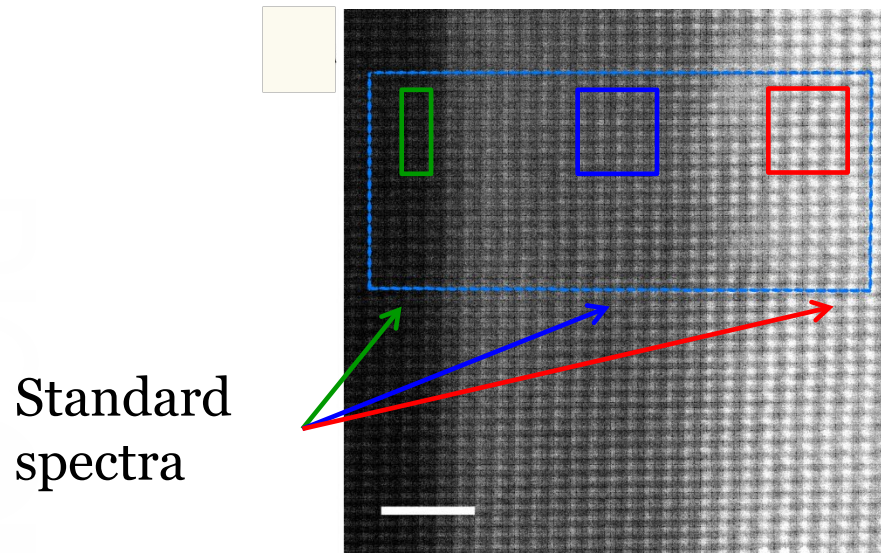


Figure by
S. Jesse

Multiple Linear Regression

Linear mixing $S(\mathbf{x}, \mathbf{R}) = \sum_i a_i(\mathbf{x}) w_i(\mathbf{R}) + N$ but $w_i(\mathbf{R})$ are **known**

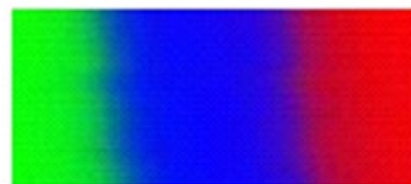
STEM of STO/LSMO/BFO interface Low-loss EELS spectra of three components



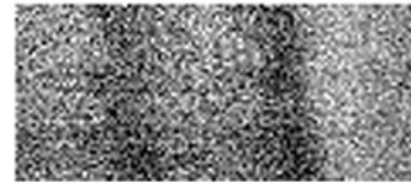
A.Y. BORISEVICH ET AL,
Suppression of Octahedral Tilts and Associated Changes in Electronic Properties at Epitaxial Oxide Heterostructure Interfaces, Phys. Rev. Lett. **105**, 087204 (2010).

“Chemistry”:
35 to 125 eV

Fit coefficient map



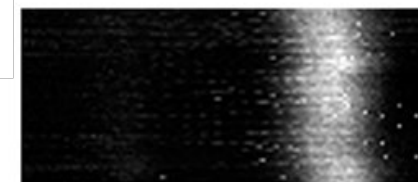
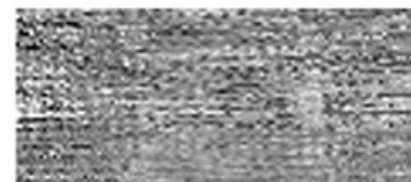
residuals map



χ^2 map



“Plasmons”
5 to 35 eV



Eigenvectors and loadings

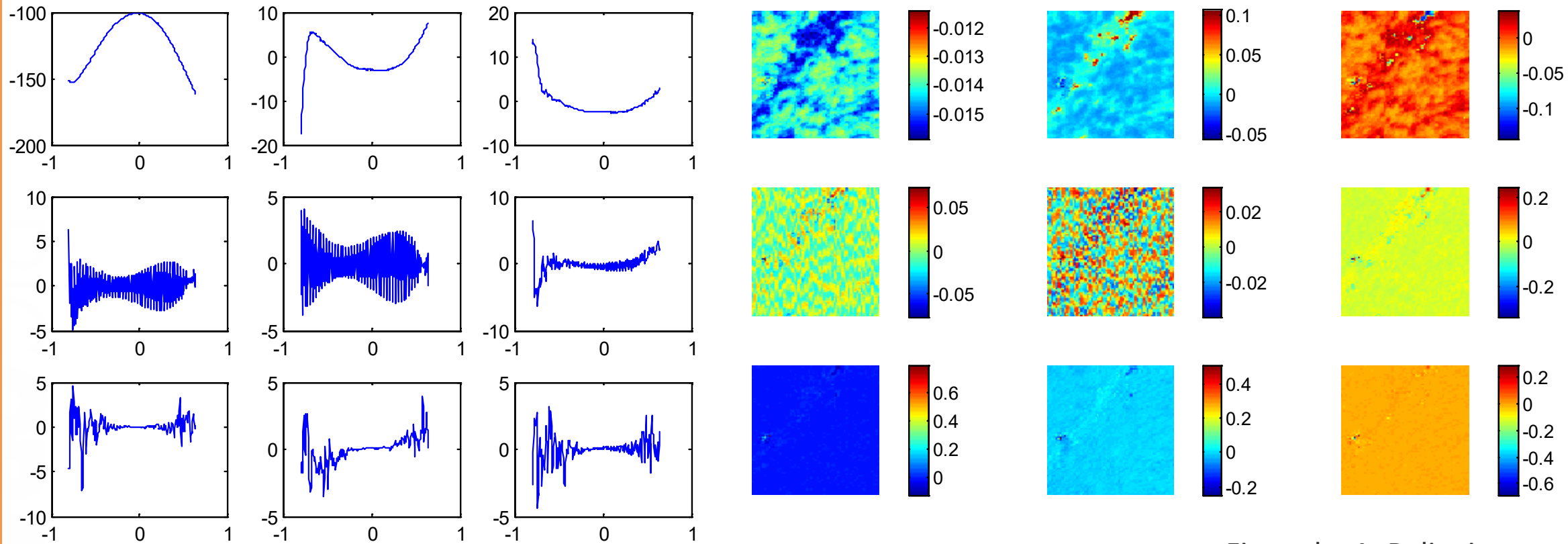


Figure by A. Belianinov

Scree plot and correlations

- Semi log plot indicating the “weight” of each component as a function of all components
- Only the first few components contain useful info, while others are dominated by noise

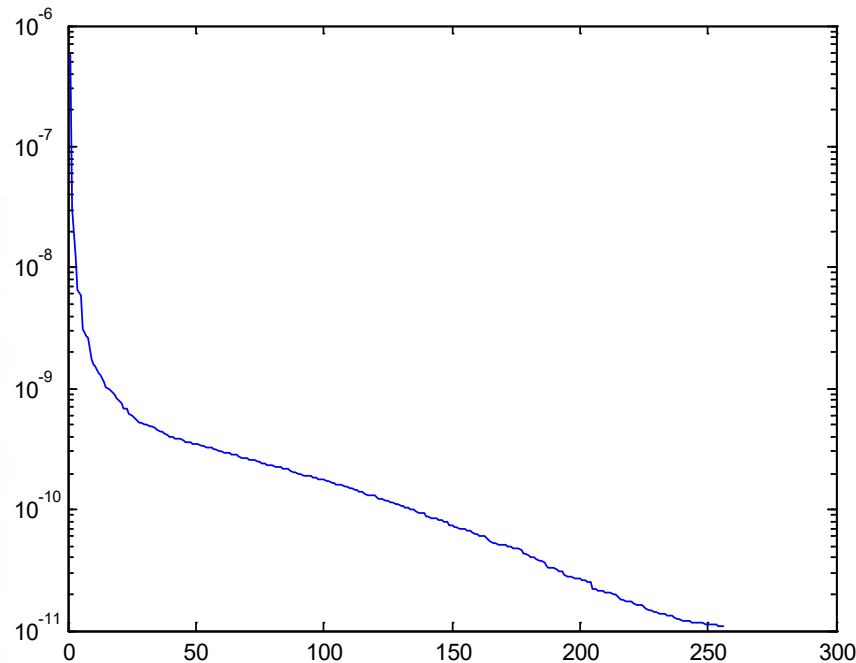
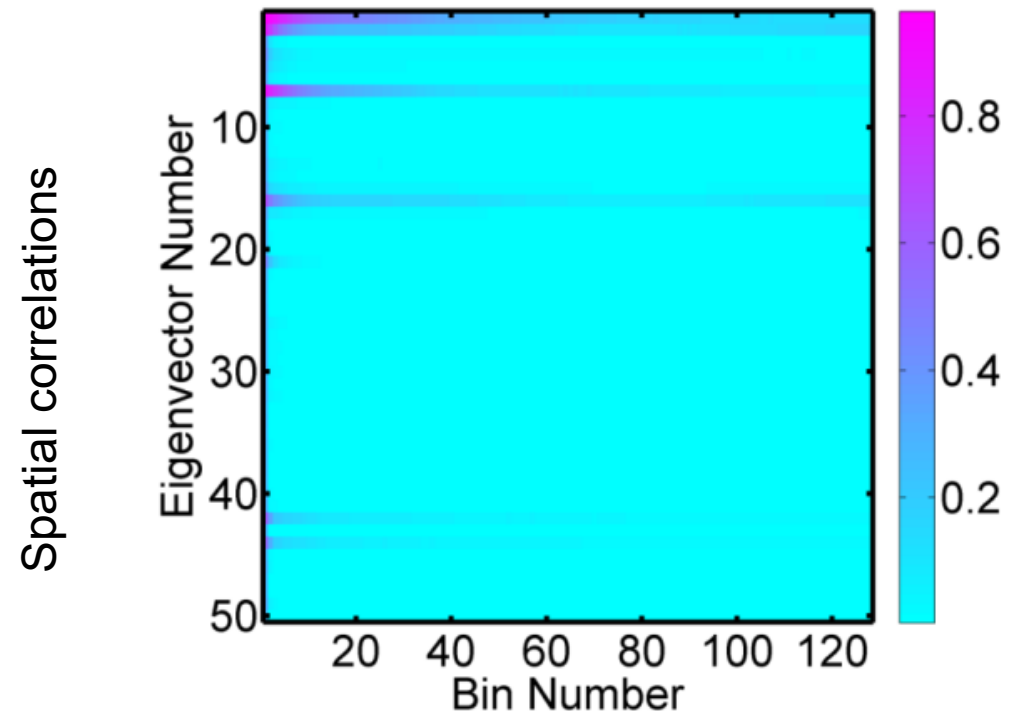


Figure by A. Belianinov

- We can also analyze correlations in images

For AFM data

PCA Eigenvectors

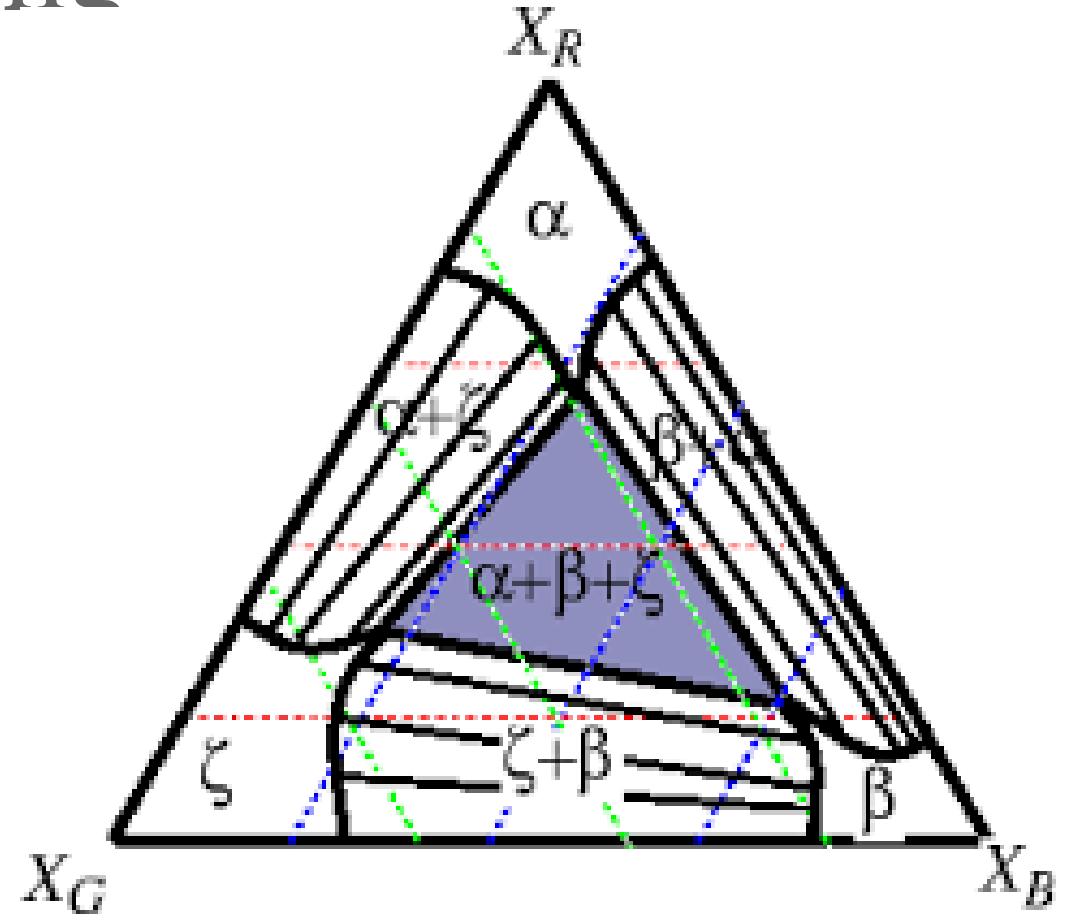


Bayesian Linear Unmixing

$$S(\mathbf{x}, \mathbf{R}) = \sum_{i=1}^K a_i(\mathbf{x})w_i(\mathbf{R}) + N$$

$$\sum_{i=1}^K a_i(\mathbf{x}) = 1$$

- The eigenvectors $w_i(\mathbf{R})$ are non-negative, $w_i(\mathbf{R}) \geq 0$
- The loading coefficients sum to 1
- The number of eigenvectors, K , is a priori unknown

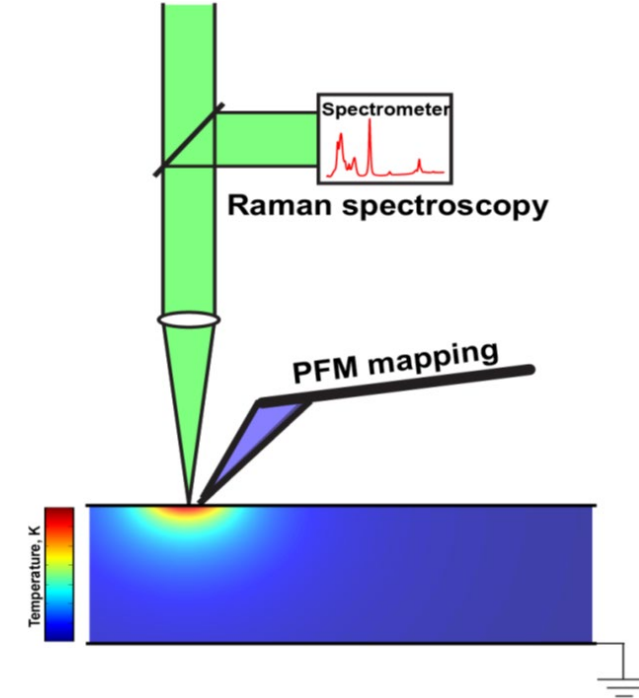


BLU is ideally suited for certain classes of problems, e.g. conduction through parallel channels, optical or electronic spectra of mixtures, etc

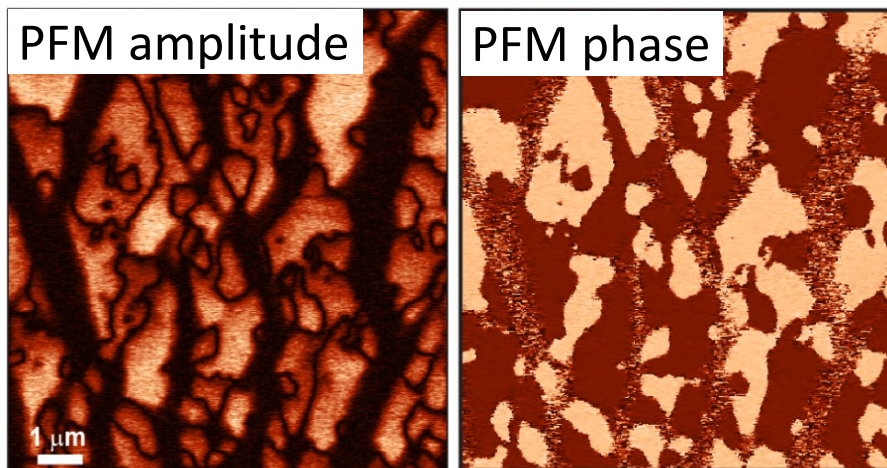
Laser heating induced phase transitions

- Copper indium thiophosphate ($\text{Cu}_{0.77}\text{In}_{1.12}\text{P}_2\text{S}_6$) layered ferroelectric
 - Ferroelectric state at room temperature
 - Curie temperature $T_c = 320$ K
 - Non-polar $\text{In}_{4/3}\text{P}_2\text{S}_6$ inclusions
- Combined Atomic Force Microscopy (AFM) and confocal Raman spectroscopy investigative approach
 - AFM – topography measurements
 - Piezoresponse force microscopy (PFM) – static ferroelectric domain structure
 - Raman – crystallographic structure via Raman spectra

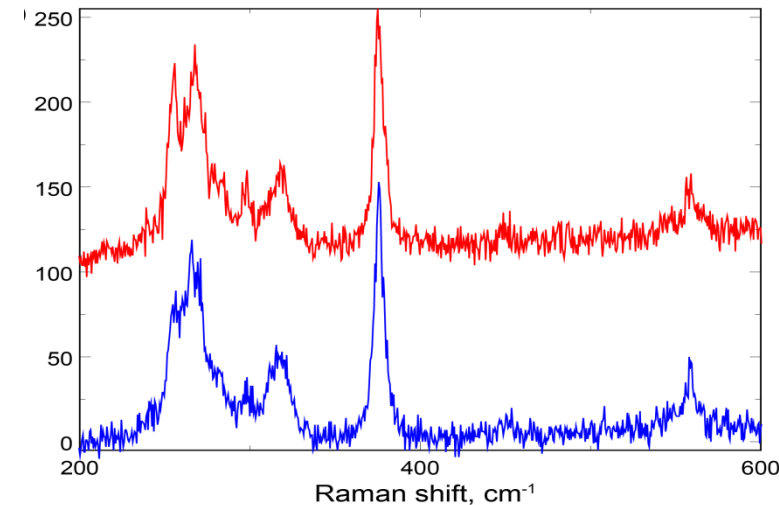
Experimental scheme



Ferroelectric domain structure



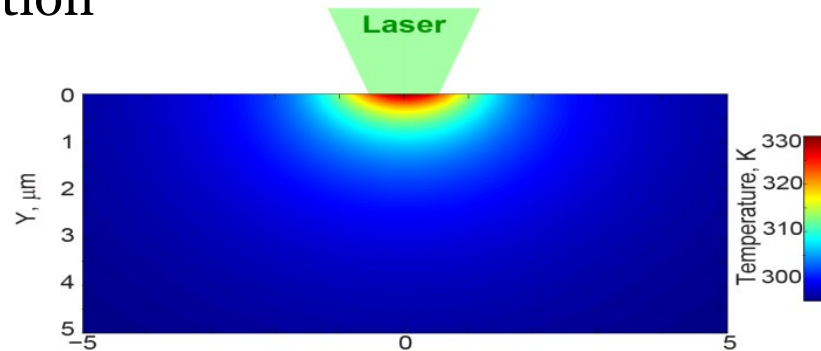
Single point Raman spectra



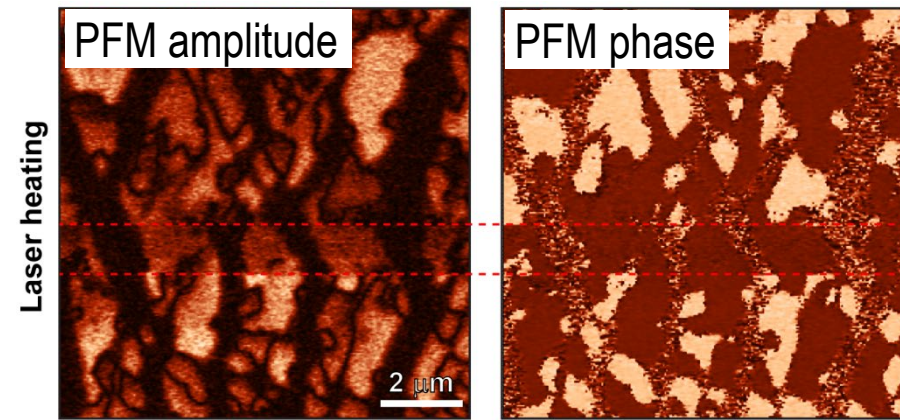
A. IEVLEV, ACS Nano
9, 12442 (2015).

Laser heating induced phase transition

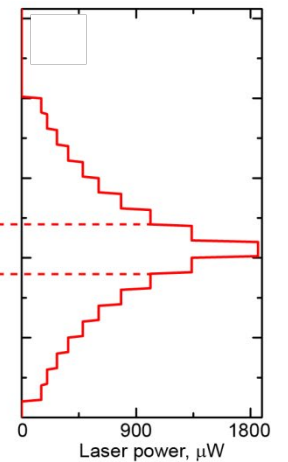
Laser can be used for local heating to induce ferroelectric- paraelectric phase transition



Domain structure evolution

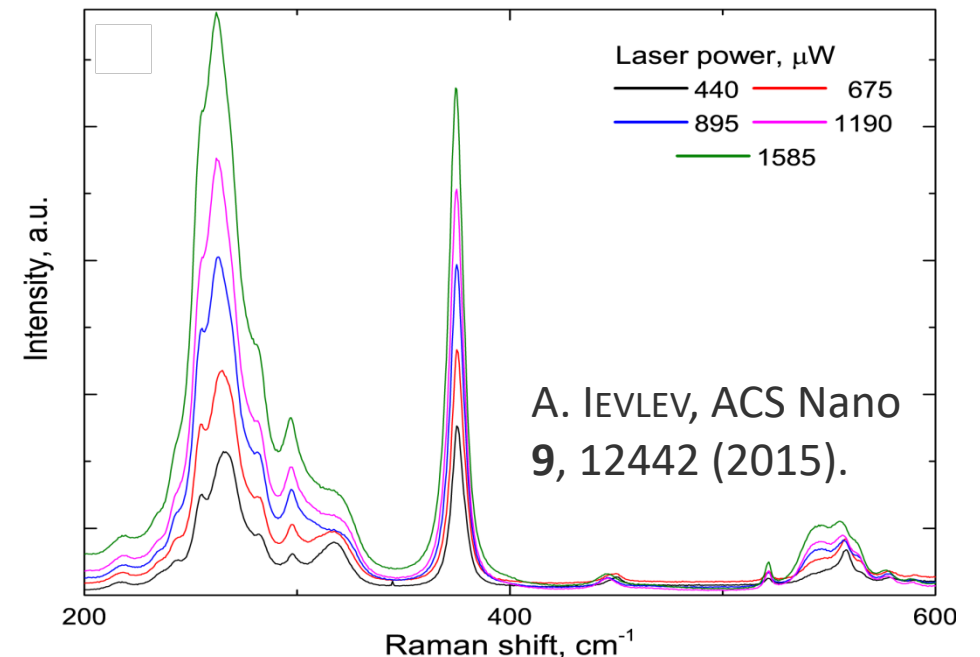


Laser power



- Measurements with variation of the laser power
 - PFM – *in-situ* change in the domain structure above T_c
 - Raman – evolution of the Raman spectra through the phase transition
- Comprehensive analysis of Raman spectra is complicated by inhomogeneous chemical composition and high noise level
- Bayesian Linear Unmixing can be used for automated identification of spectra evolution

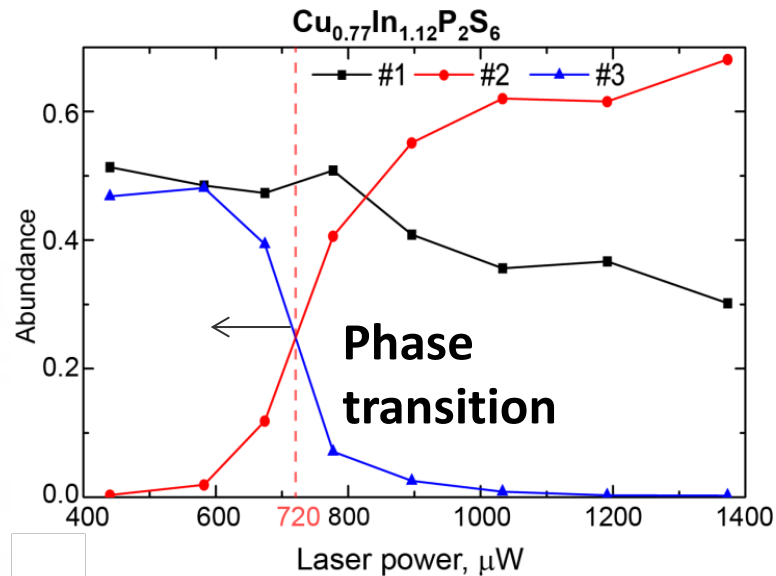
Raman spectra evolution (averaged)



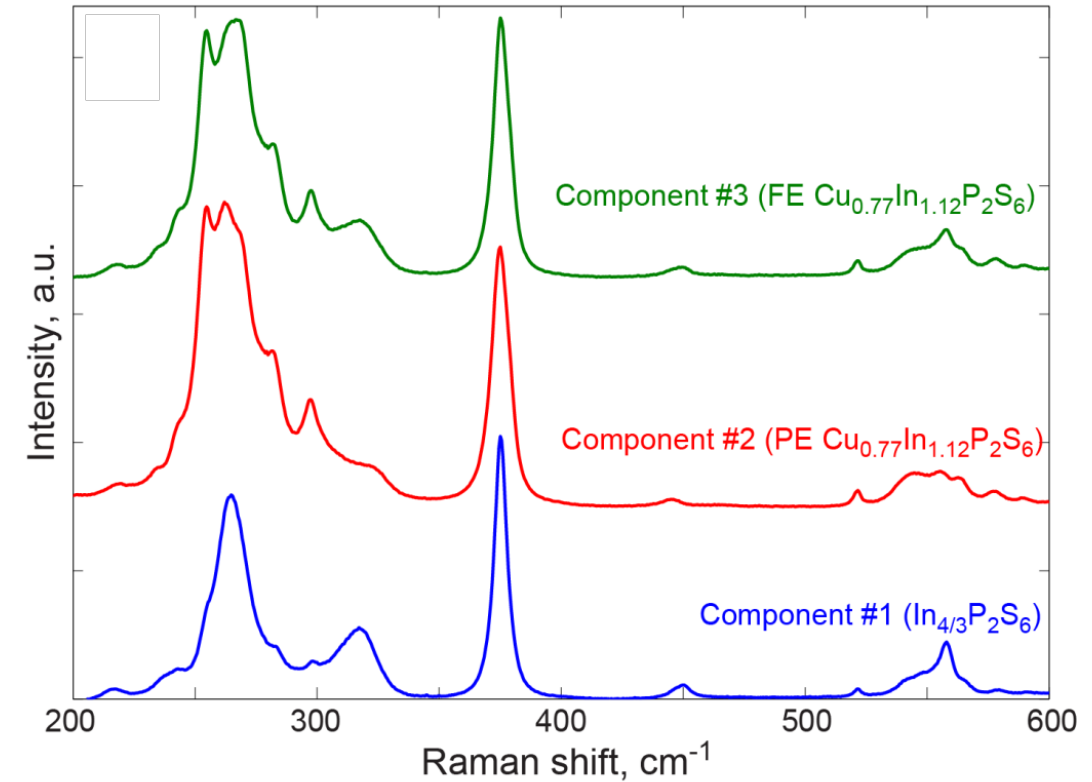
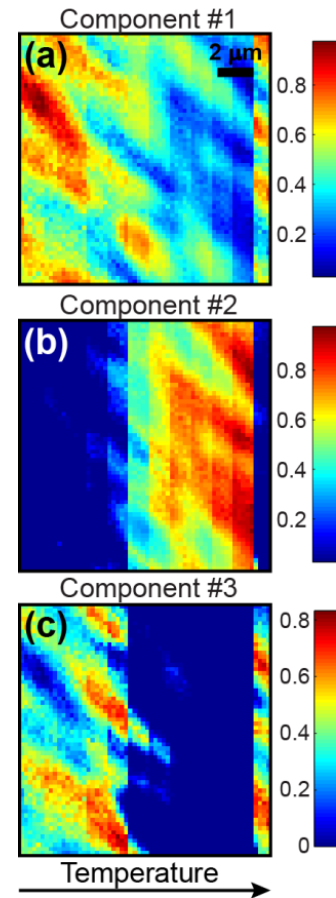
BLU separation of components

Spatial concentration of components

Results of BLU: components and loading maps



A. IEVLEV, ACS Nano **9**, 12442 (2015).

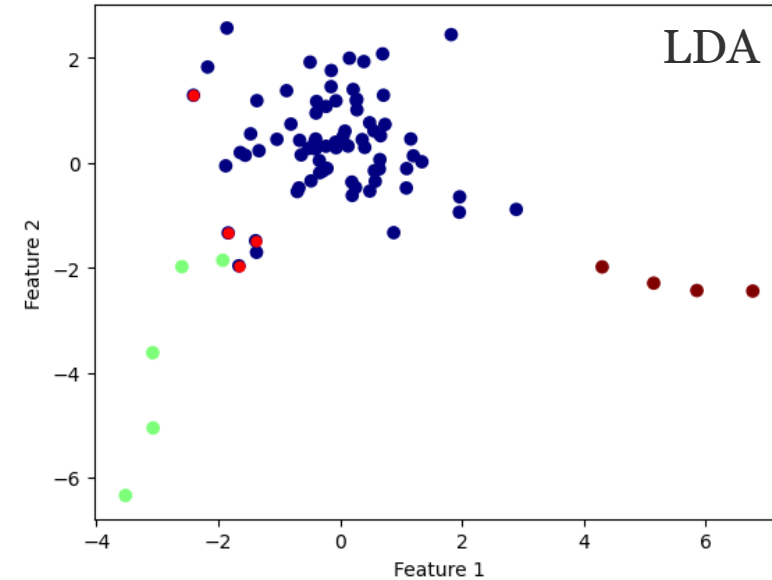
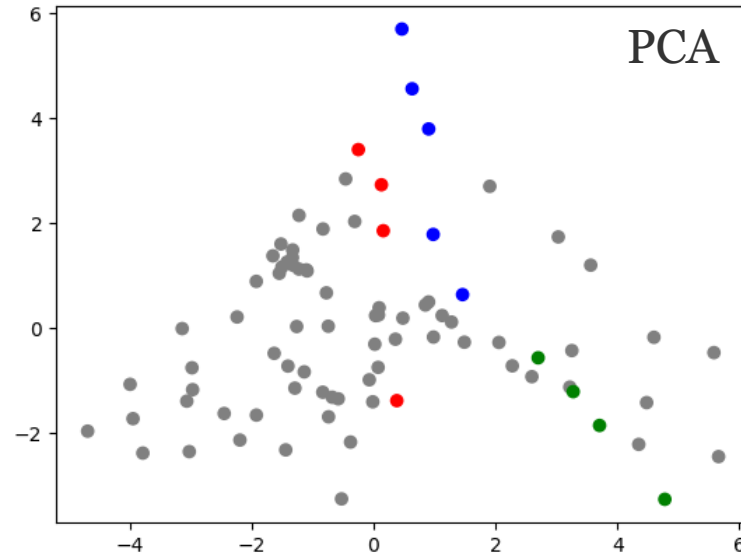


Unmixing showed presence of three independent components in Raman spectra:

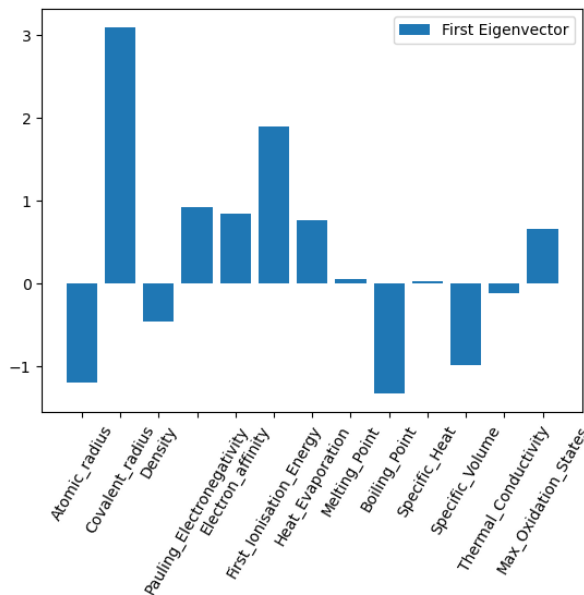
1. Non-polar $\text{In}_{4/3}\text{P}_2\text{S}_6$ – weak changes in intensity with temperature
2. Paraelectric CuInP_2S_6 above T_c – appears at higher laser powers
3. Ferroelectric CuInP_2S_6 below T_c – disappears at higher temperatures

PCA vs. LDA for elements

Alkali
Alkali
earth
Halogens

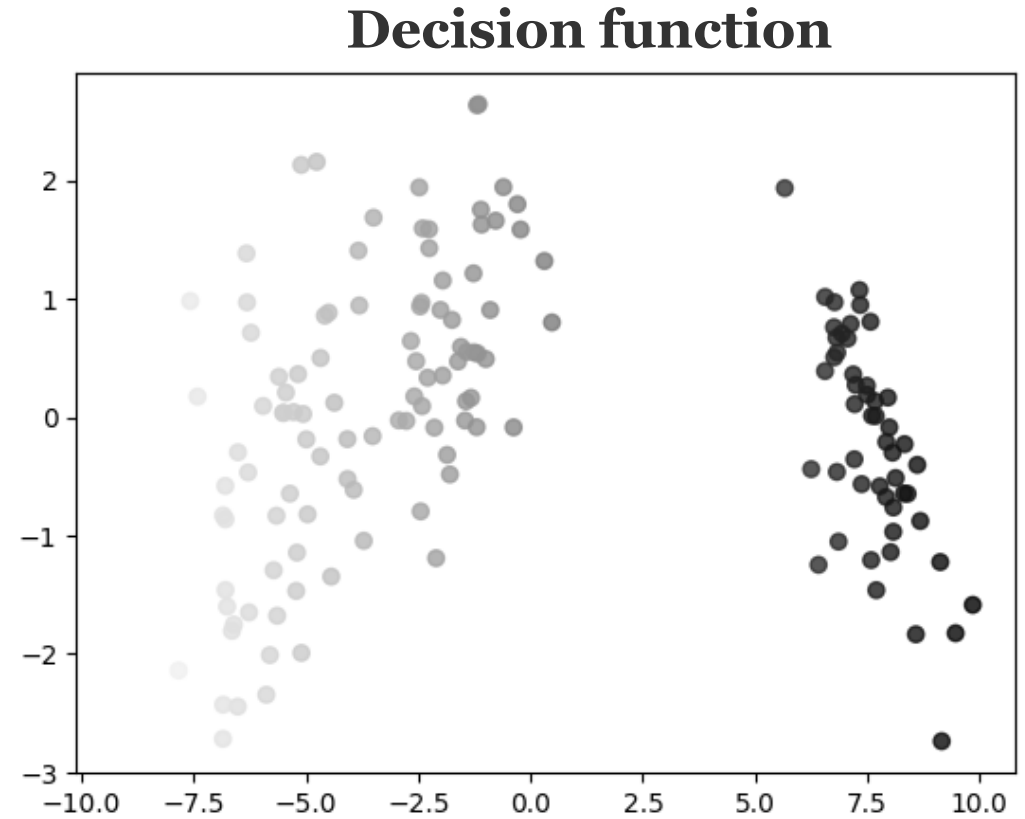
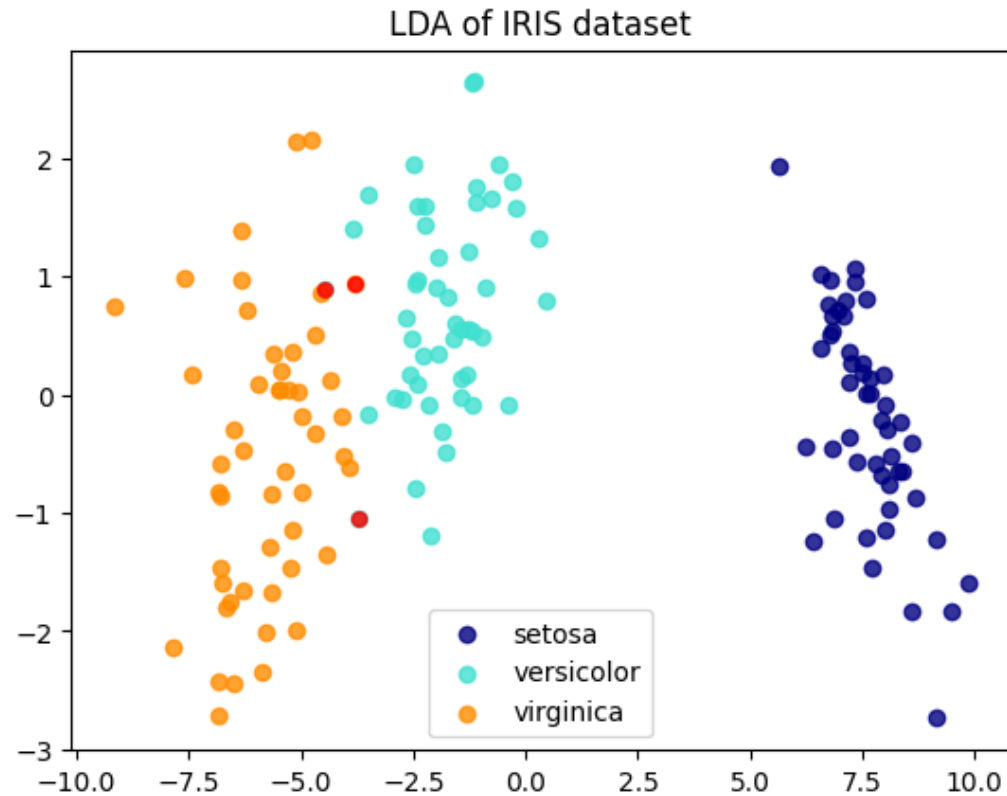


Alkali
Alkali
earth
Halogens



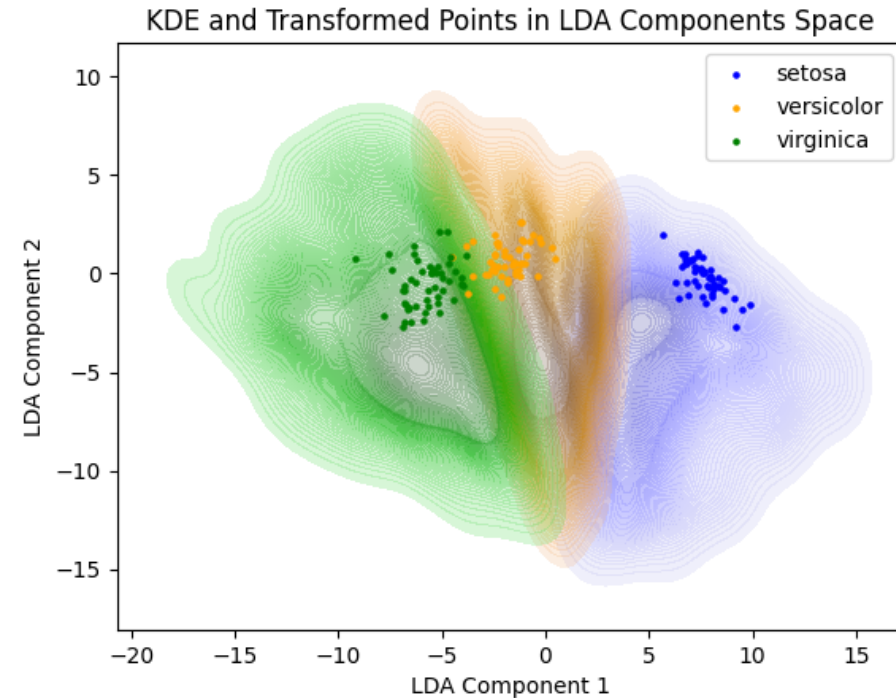
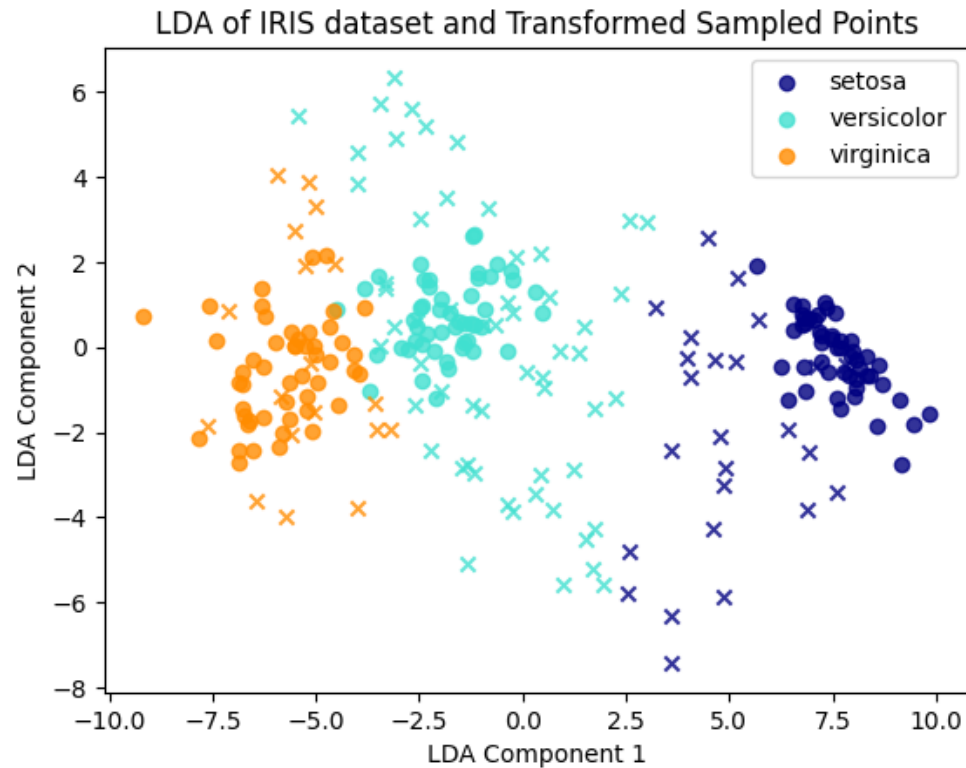
- Our element space is described by 13 descriptors
- In PCA, we found 2 linear combinations of these descriptors that describe this data set best.
- Alkali, alkali-earth, and halogens are close to each other in PCA space
- LDA finds best representation to separate alkali and halogens from everything else
- In LDA representation, alkali earth are close to alkali

What else can LDA give us?



- Decision function allows us to quantify how likely is the feature to belong to certain class

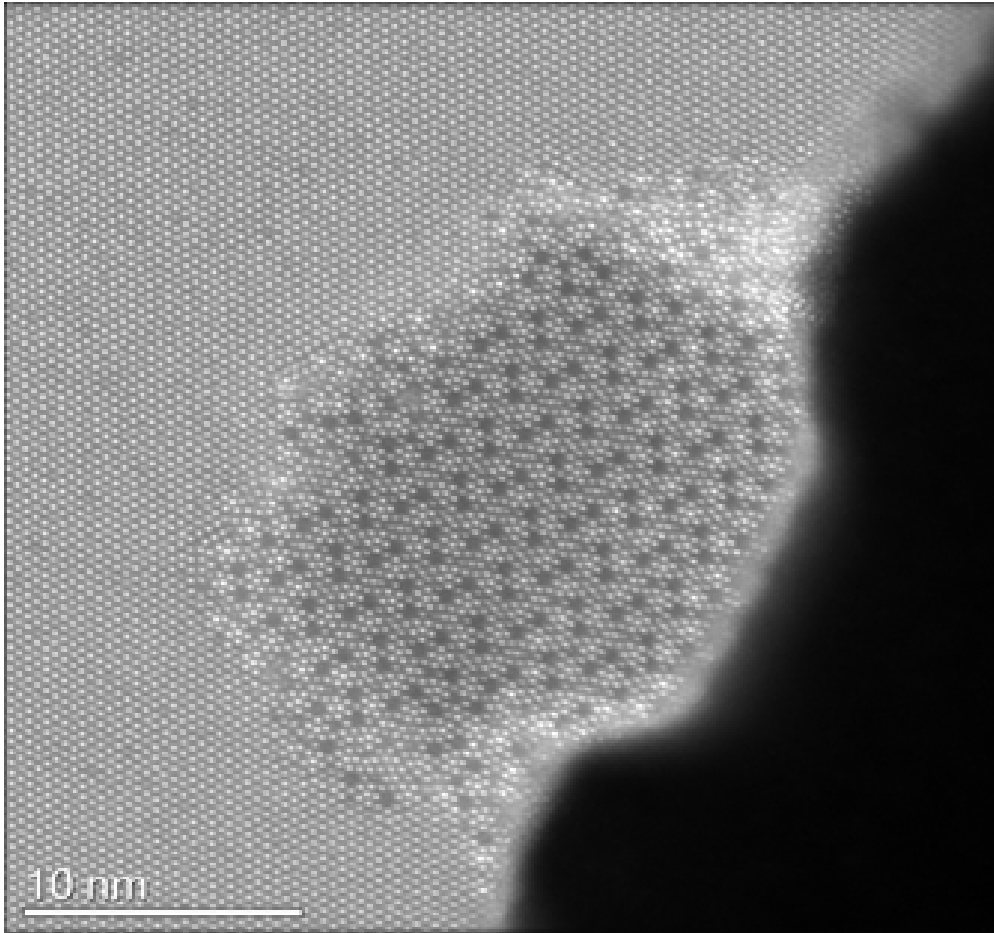
Visualizing the decision surfaces



- Generate multiple points uniformly distributed in the original high-dimensional space
- Perform the LDA transform
- Calculate the KDE

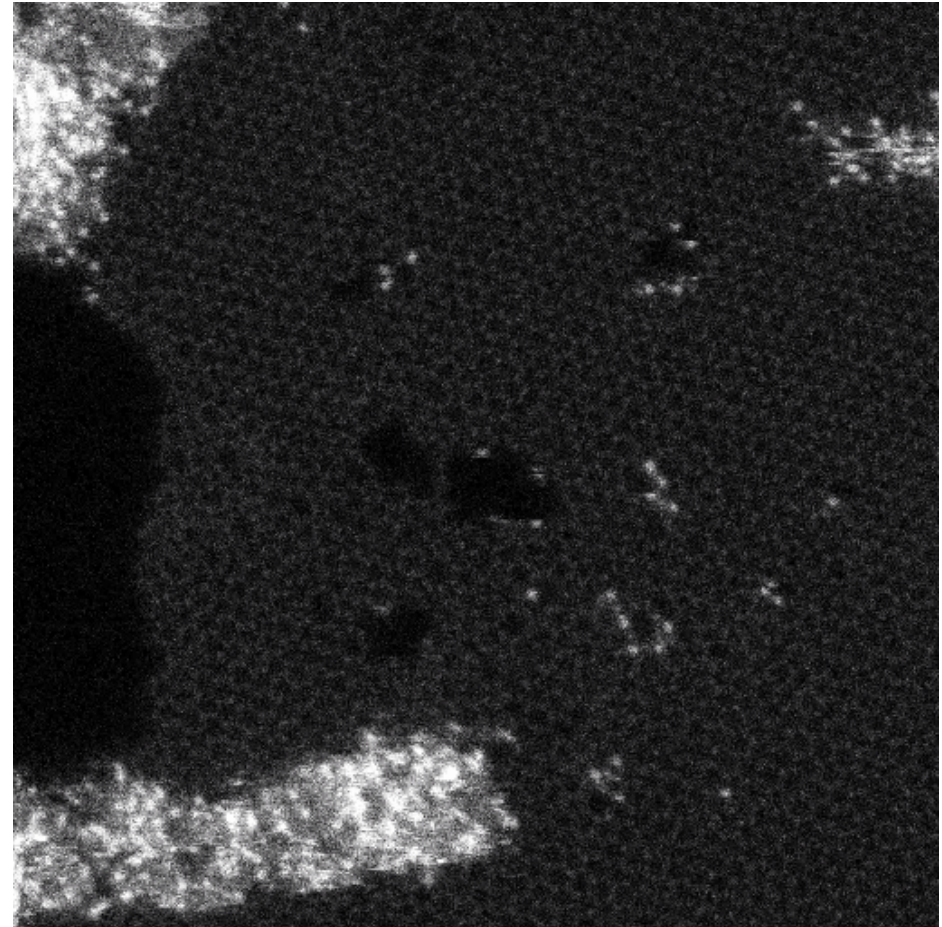
Chemically disordered systems

Mo-V-Ta complex oxide



Q. He et al, ACS Nano 9, 3470-3478

Si in graphene

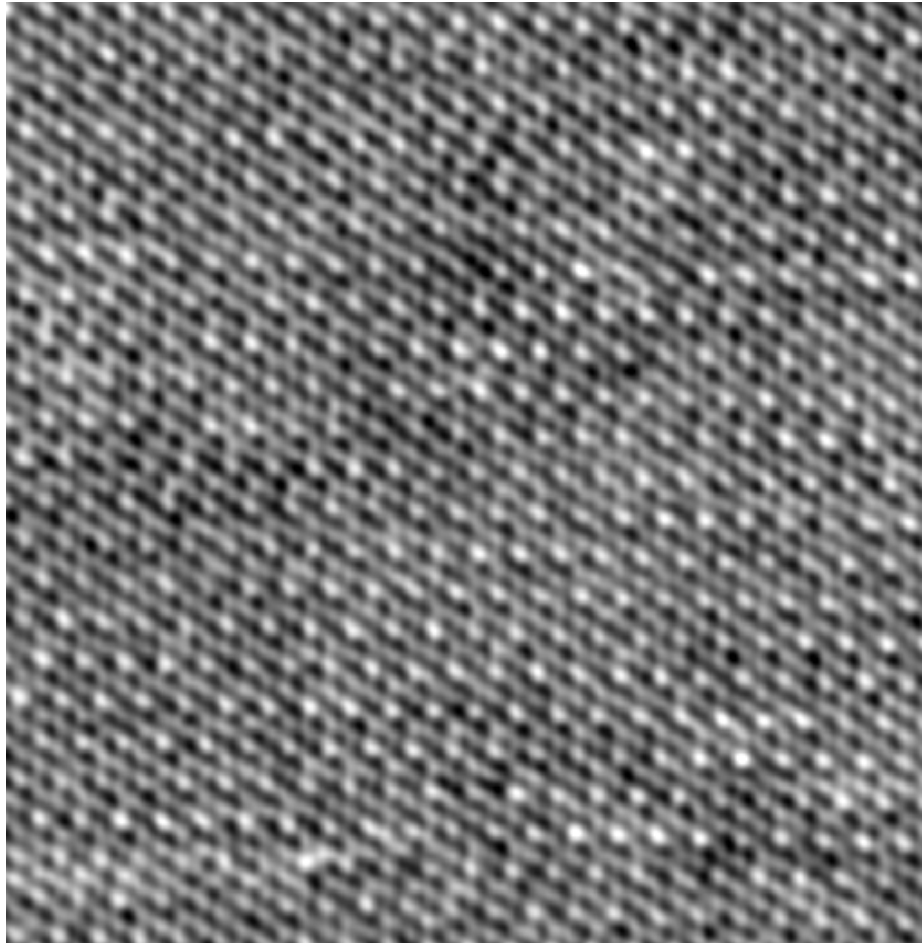


Data collected by O. Dyck (ORNL)

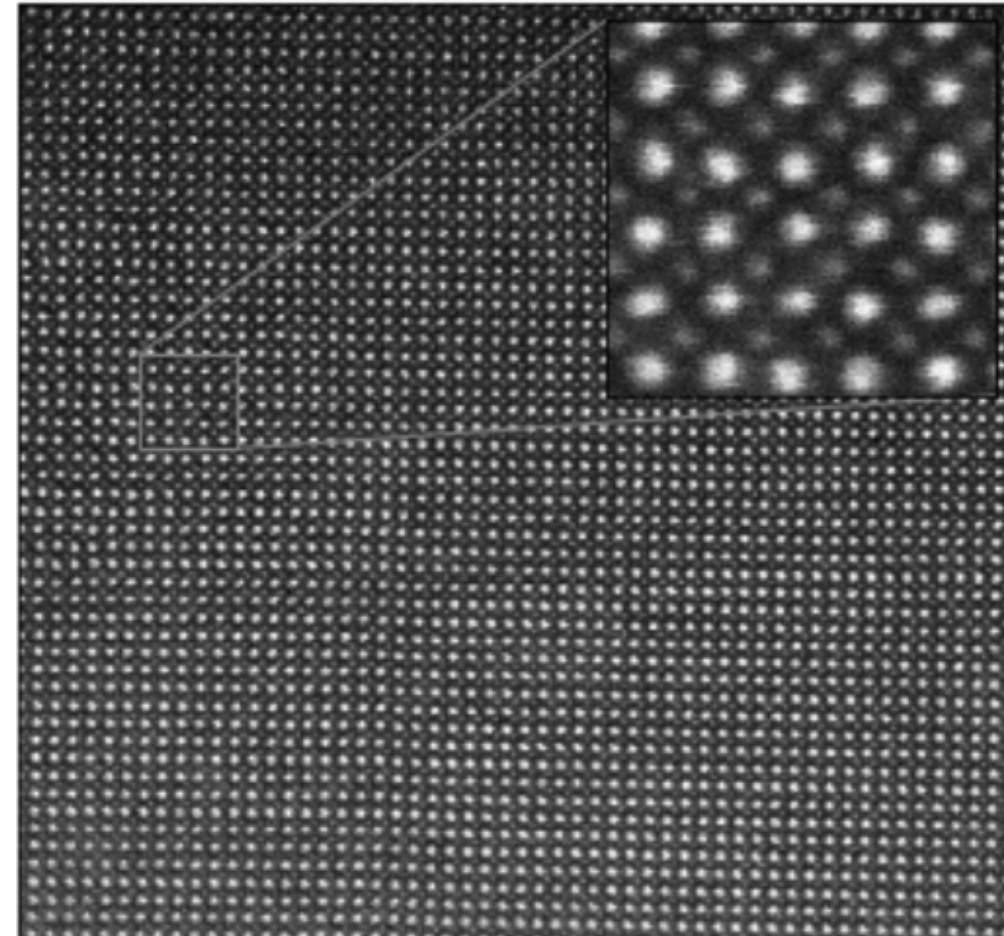
- What is the nature of the building blocks and relevant atomic configurations?
- Can we define single-phase regions and phase boundaries?

But what about subtle distortions?

Electronic structure in RuCl_3



BiFeO_3 on SrRuO_3



- Can we identify ferroelectric and ferroic variants and associated topological defects?
- What is the nature of the phases?

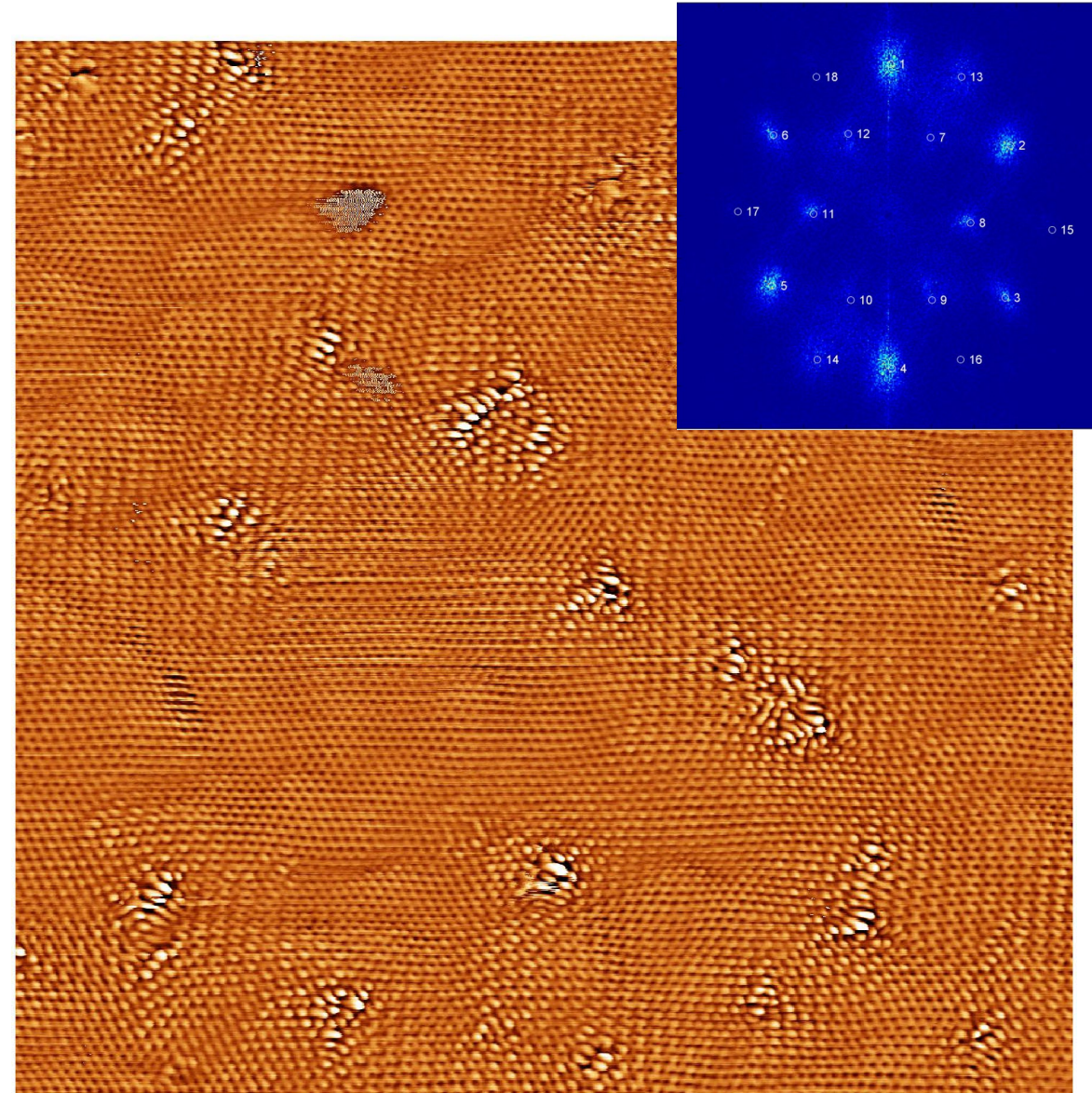
Can global FFT help?

- Global FFT – everything is averaged:
 - Drift
 - Extended defects
 - Multiple grains

We are averaging out all interesting phenomena except for small spatially uniform structural distortions

- Solution – sliding window approaches:
 - Fit FFT peaks: amplitudes, positions
 - Multivariate analysis

Note: window can be also tied to a specific feature, such a selected atom. Then we explore atomic neighborhood



General linear unmixing

$$S(\mathbf{x}, \mathbf{R}) = \sum_i a_i(\mathbf{x}) w_i(\mathbf{R}) + N$$

We start with:

- $\mathbf{x}$ is the spatial variable, $\mathbf{x} = (x, y)$
- $\mathbf{R}$ is the (vector) parameter variable

We aim to get:

- $a_i(\mathbf{x})$ are loading maps
- $w_i(\mathbf{R})$ are endmembers/eigenvectors
- N is noise

The M pixel 2D image is transformed to M/N pixel image of more complex structure.

Our loading map is 2D image, and endmembers/eigenvectors are 2D images

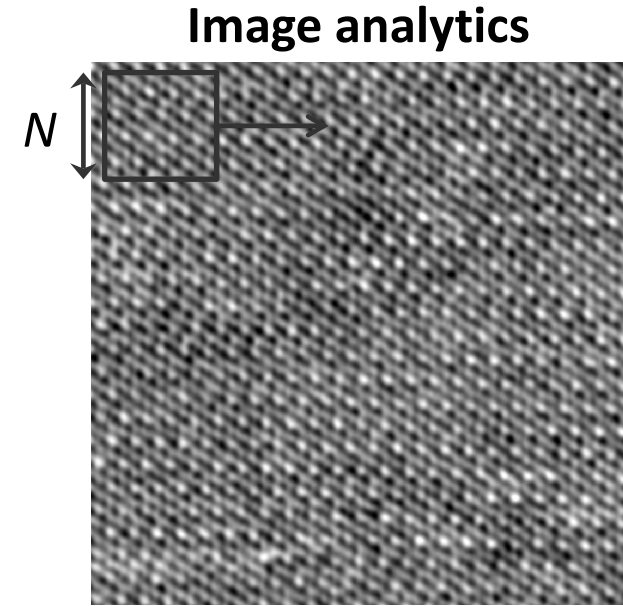
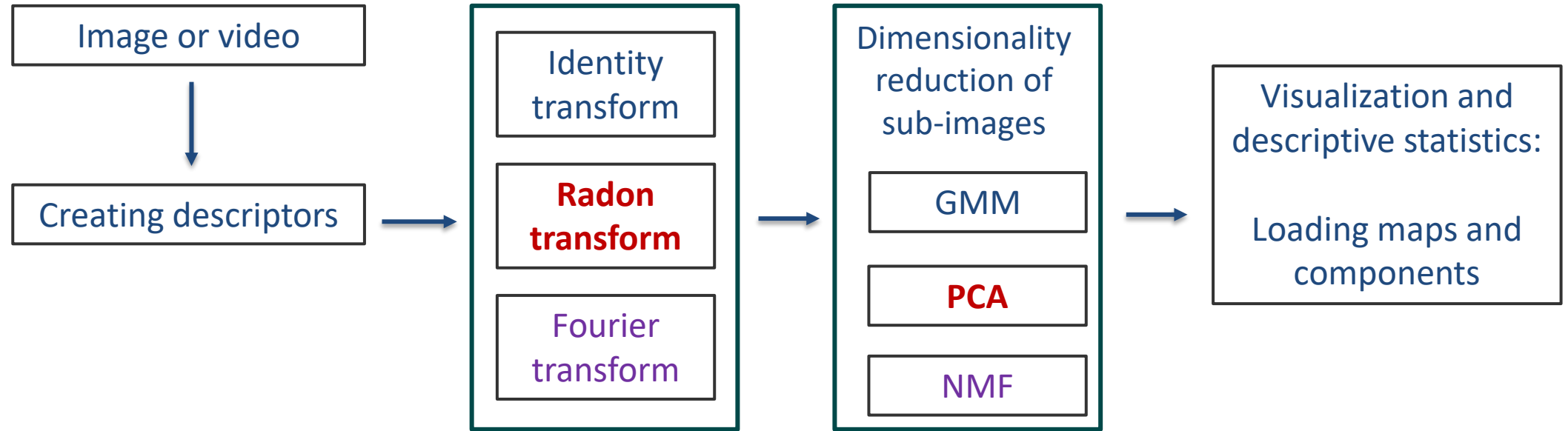


Figure by M. Ziatdinov

Sliding image transforms:

- Fast Fourier Transforms
- Correlation functions
- Intensity histograms
- Structural descriptors

Example of analysis pipeline



Pipelines are defined to

- Make analysis traceable, repeatable, explainable, and transferable
- Allow for hyperparameter tuning and optimization
- Efficiently use the memory

Sliding PCA-FFT

Can we use PCA of FFT transform in sliding windows to find periodicity?

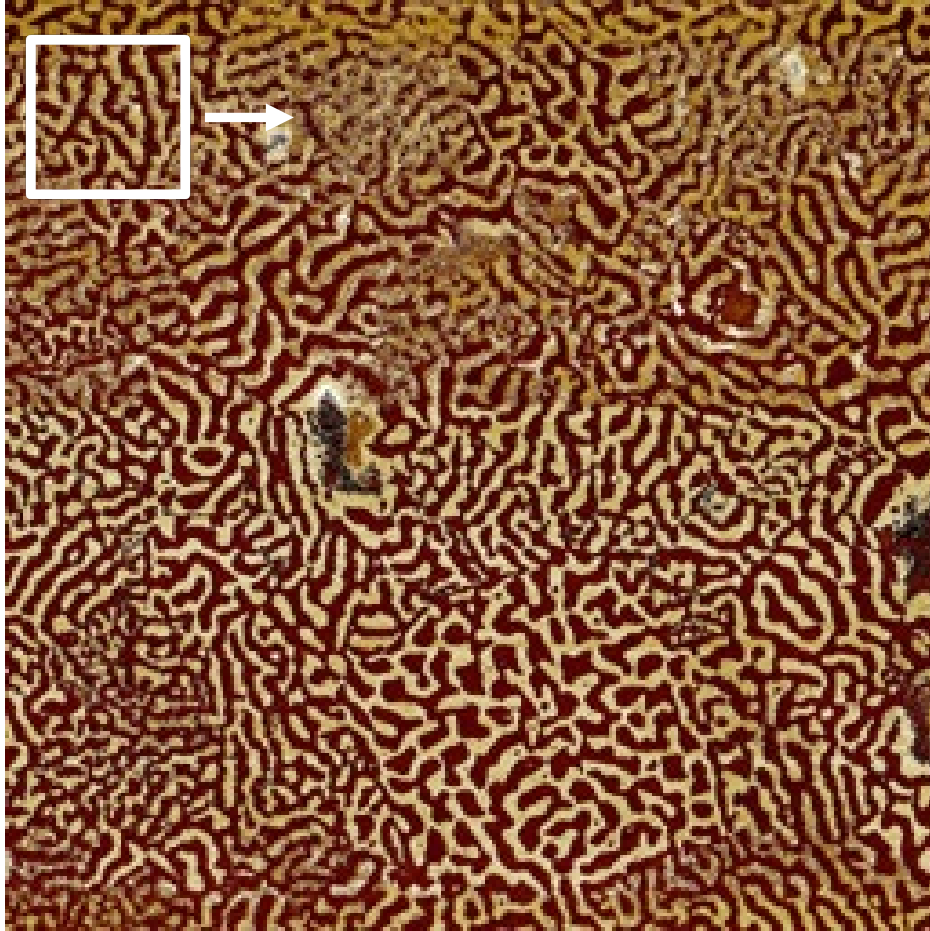
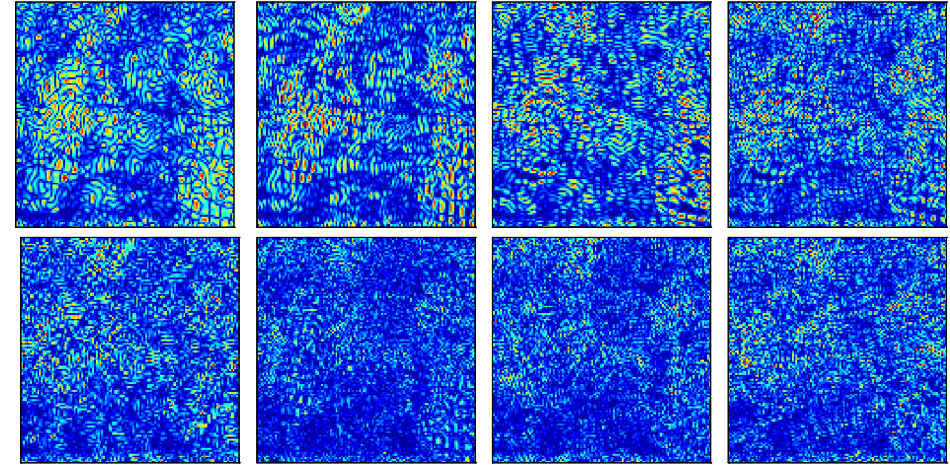
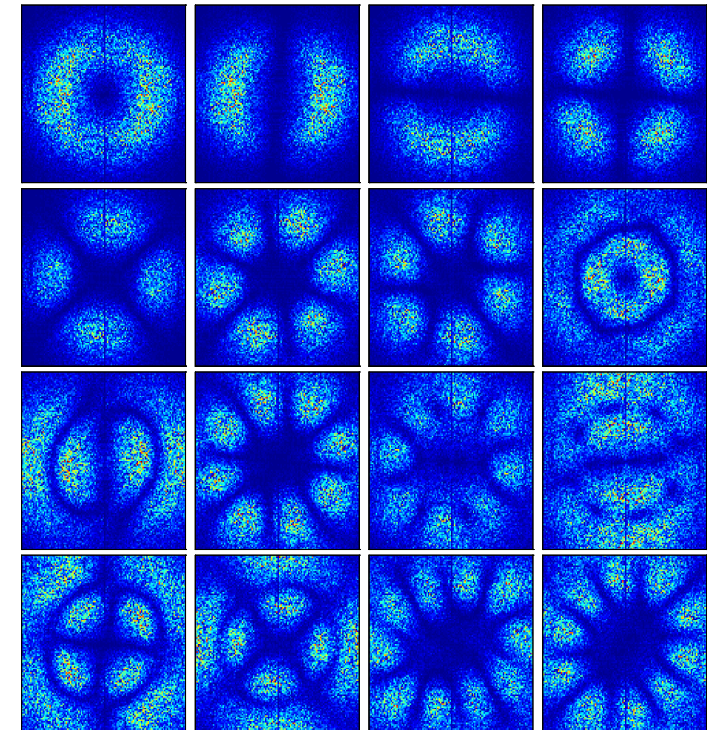


Figure by S. Jesse, data D. Gobelic

First 8 maps



First 16 eigenvectors



Spectral Unmixing: N-FINDR

Spectra for a given pixel is assumed to be a linear combination of the end-member spectra (+ Gaussian noise). The mixing proportions sum to 1

Physics constraint

$$p_{ij} = \sum_k e_{ik} c_{kj} + \varepsilon \quad \sum_k c_{kj} = 1$$

- Let E be the matrix of end-members (here, 3).

$$E = \begin{bmatrix} \mathbf{1} & \mathbf{1} & \mathbf{1} \\ \vec{e}_1 & \vec{e}_2 & \vec{e}_3 \end{bmatrix} \quad V \left(\frac{\mathbf{1}}{(l-1)!} \right) |\det(E)|$$

- Iteratively select endmembers, accepting the new selection if the volume increases

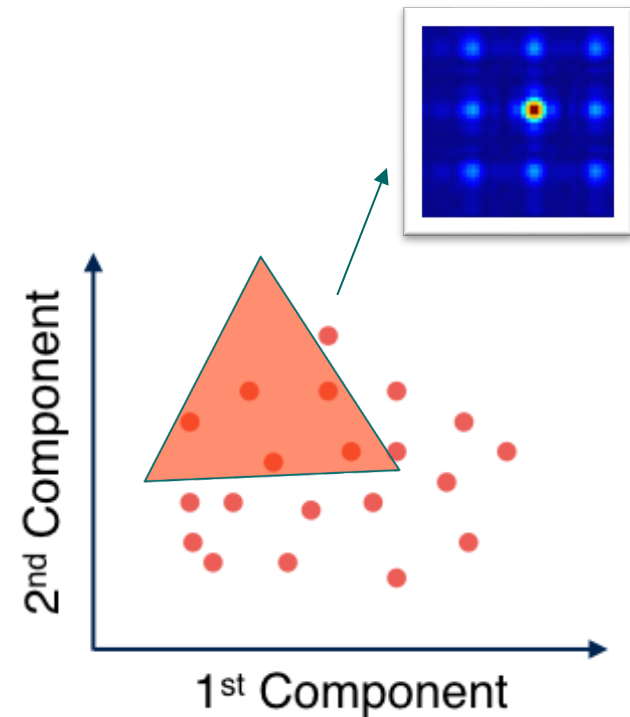


Figure by R. Vasudevan

Ideal test case

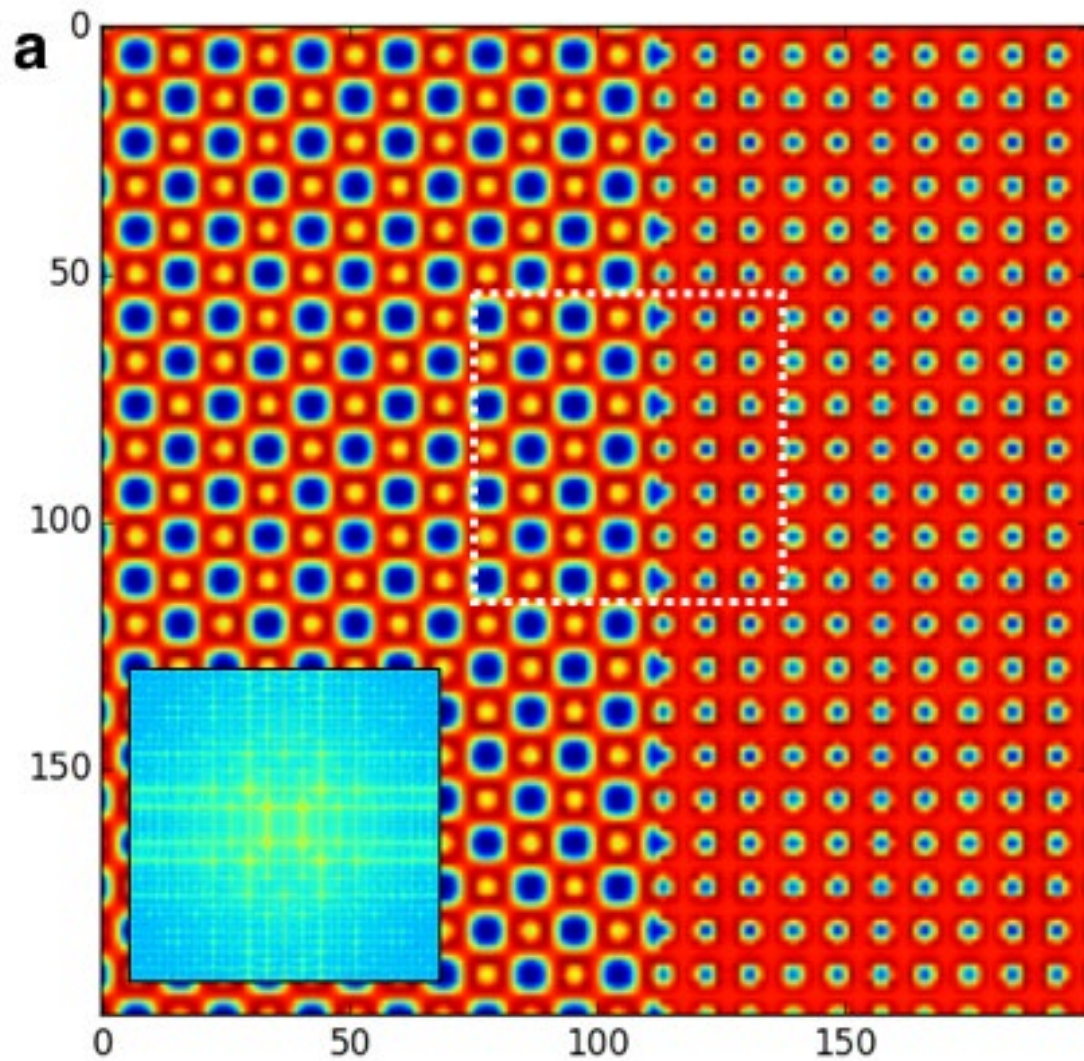
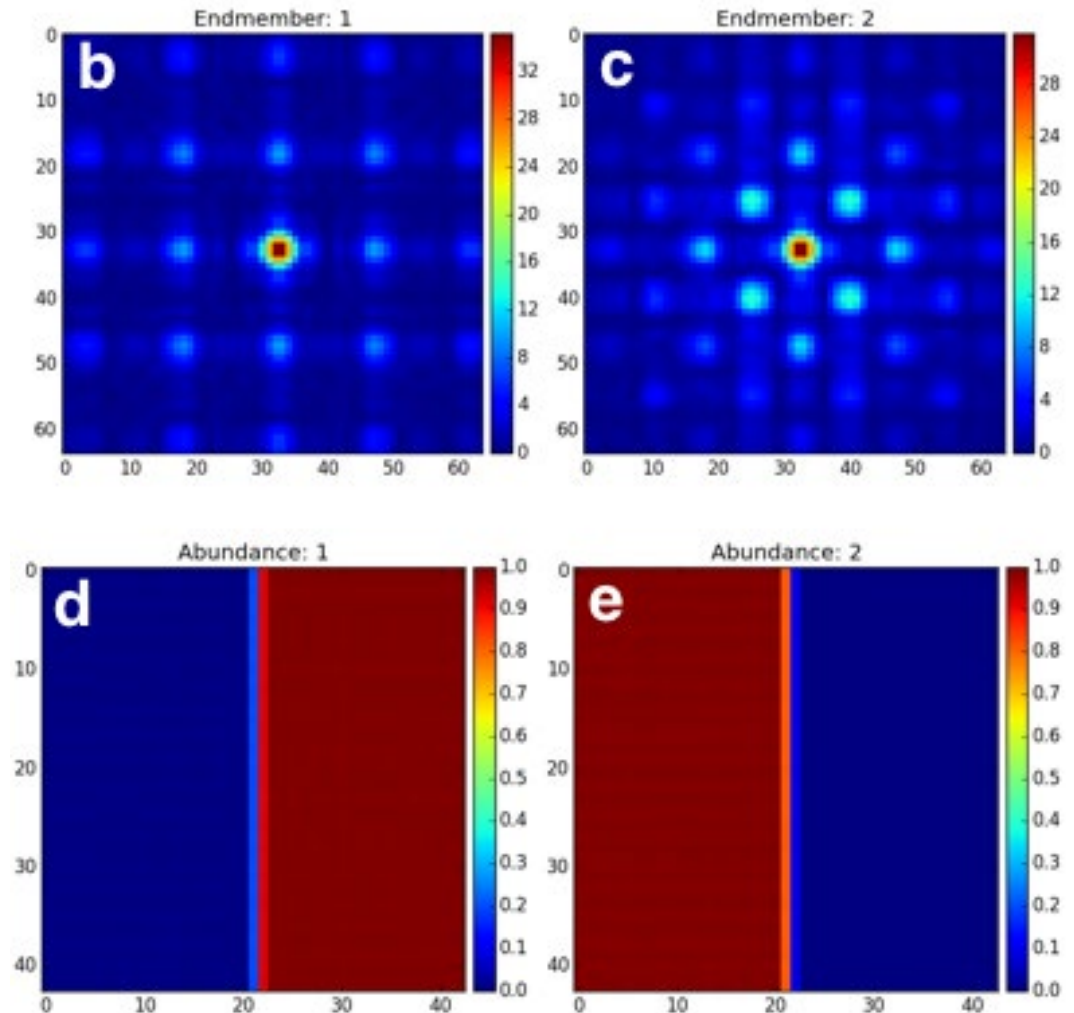


Figure by R. Vasudevan

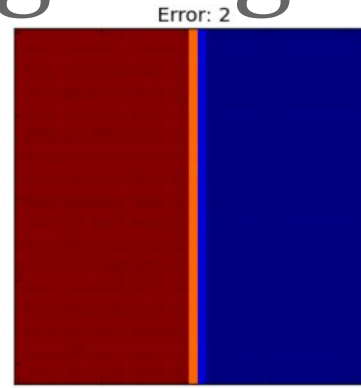
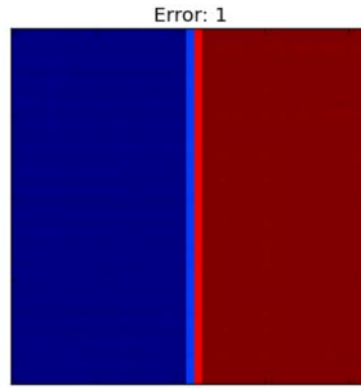


Main idea:

- FFT amplitudes are non-negative;
- FFT removes translation

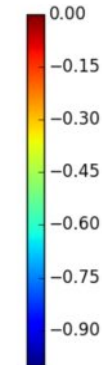
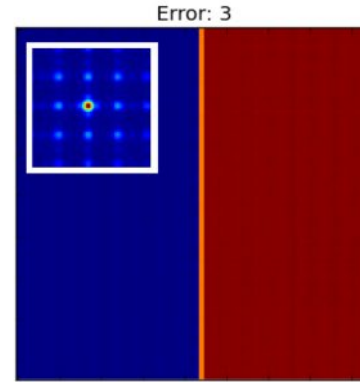
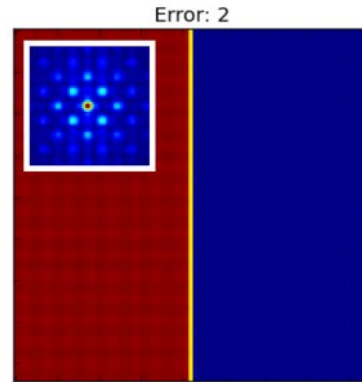
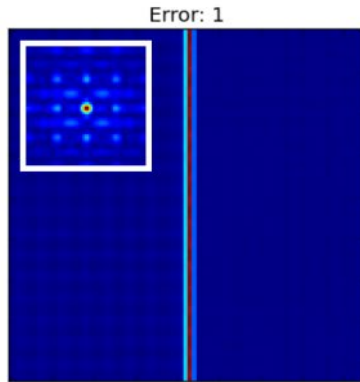
N-FINDER for image segmentation

$n_c = 2$



We can determine number of phases based on spatial localization and geometry

$n_c = 3$



$n_c = 4$

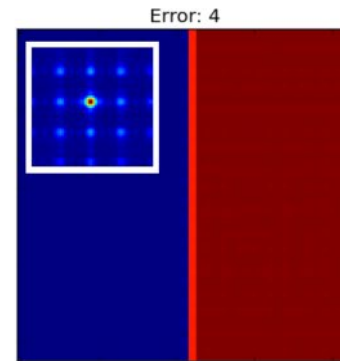
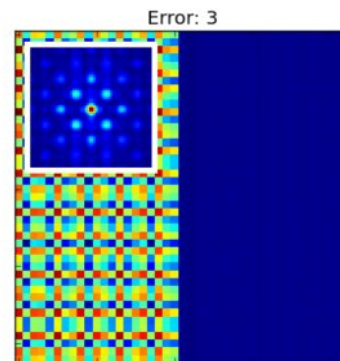
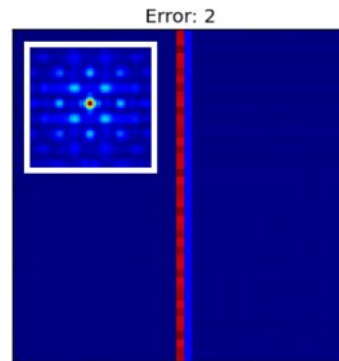
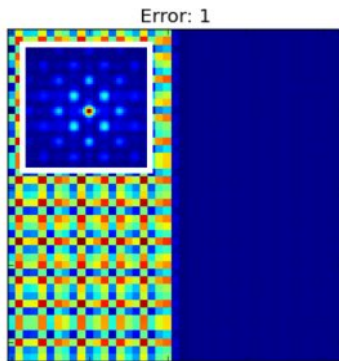
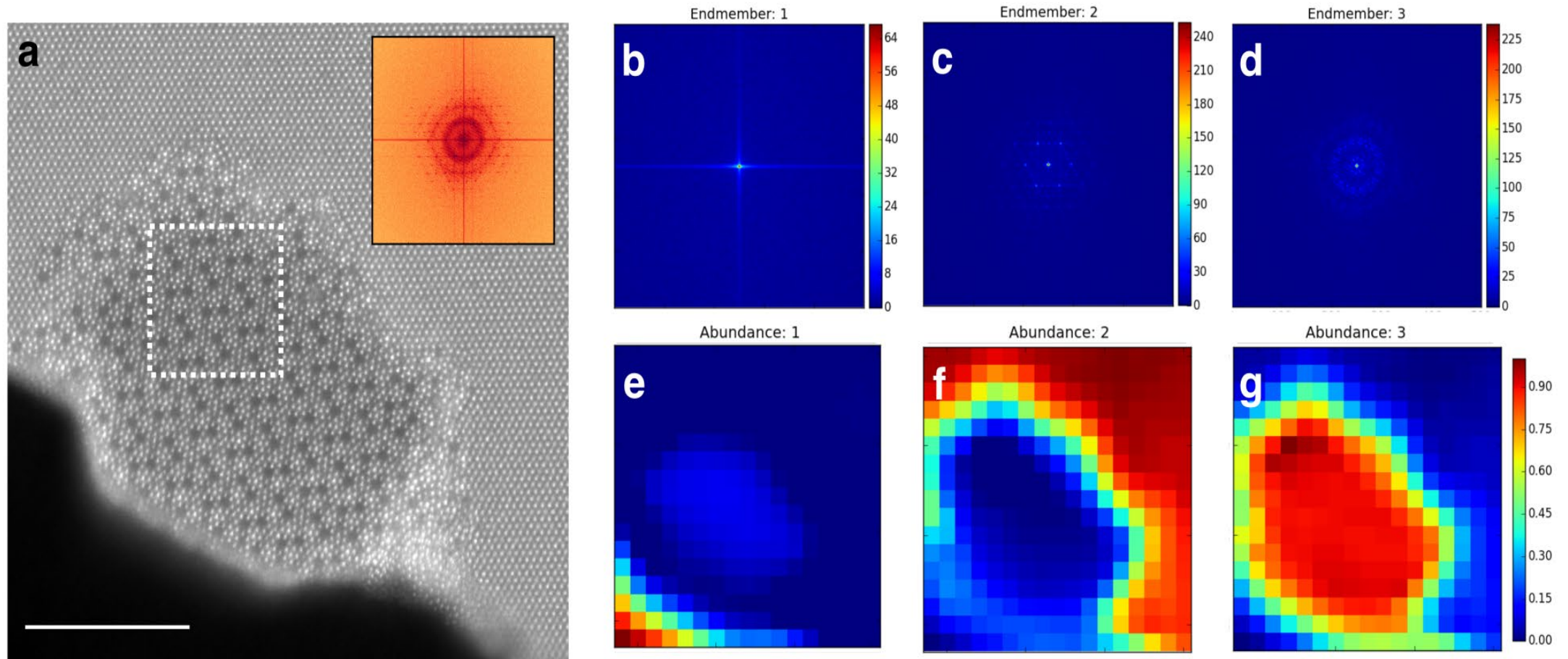
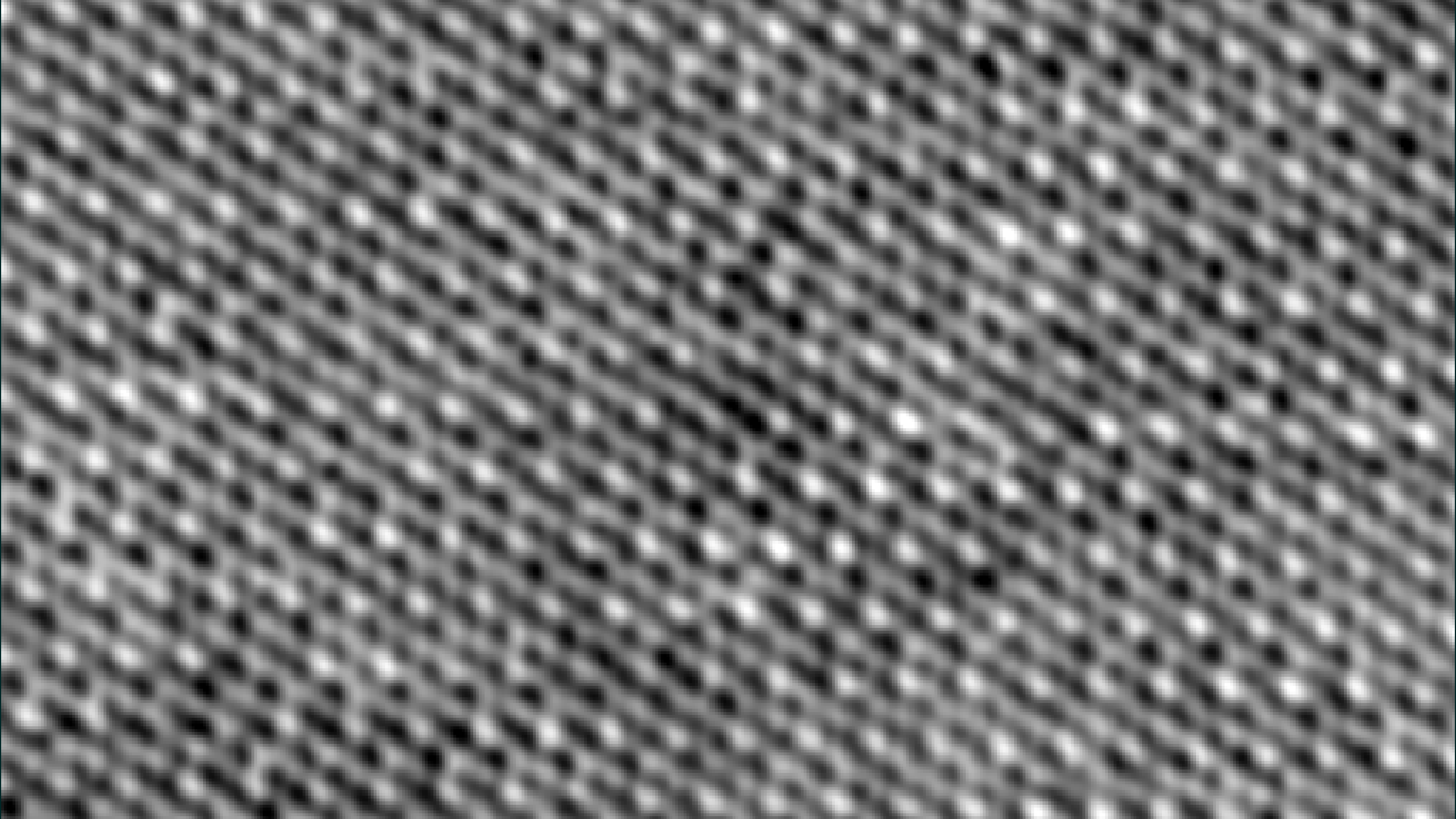


Figure by R. Vasudevan

N-FINDR for chemically separated images

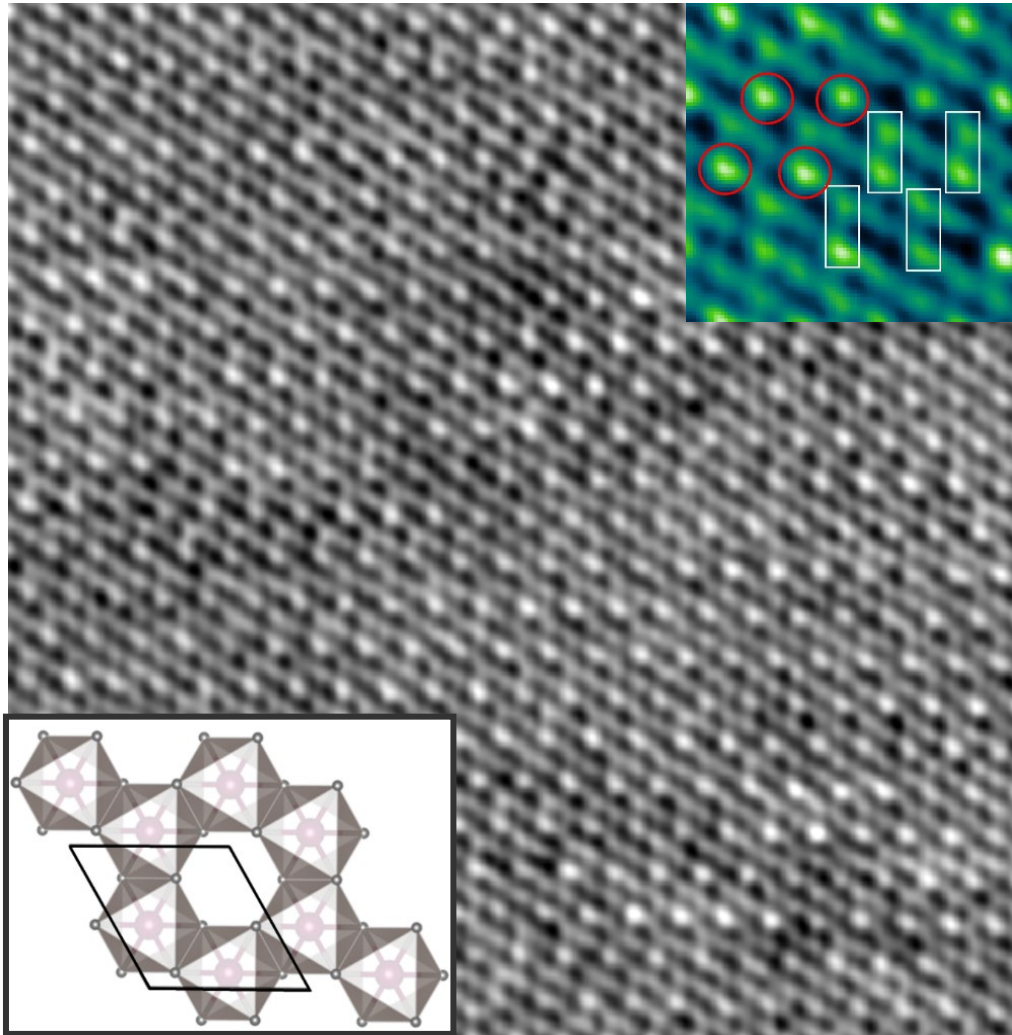


Q. He, J. Woo, A. Belianinov, V.V. Guliants, A.Y. Borisevich, *Better catalysts through microscopy: mesoscale M1/M2 intergrowth in Molybdenum–Vanadium based complex oxide catalysts for propane ammoxidation*, ACS Nano 9, 3470-3478

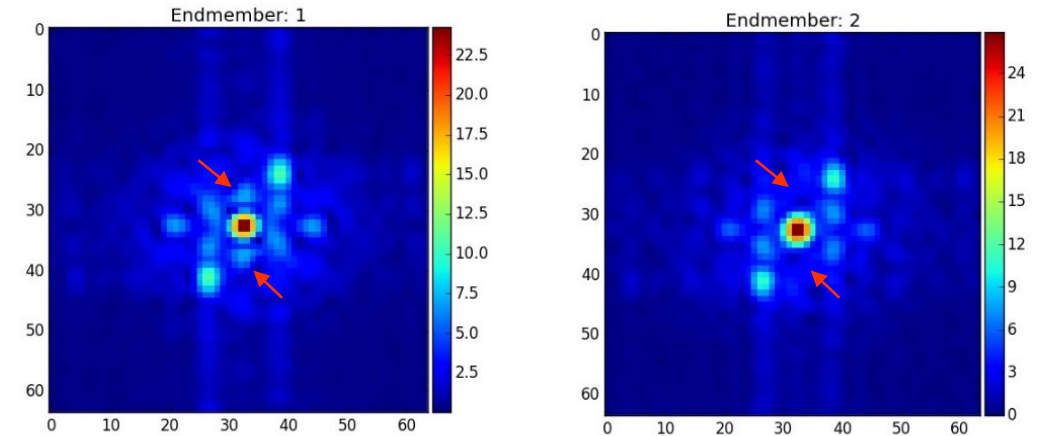


NFINDER for coexisting order parameters

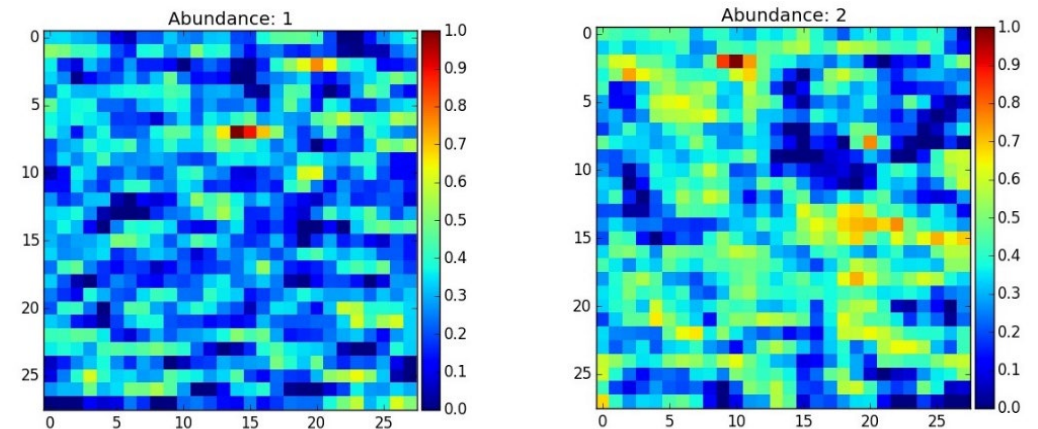
Input experimental image



FFT endmembers



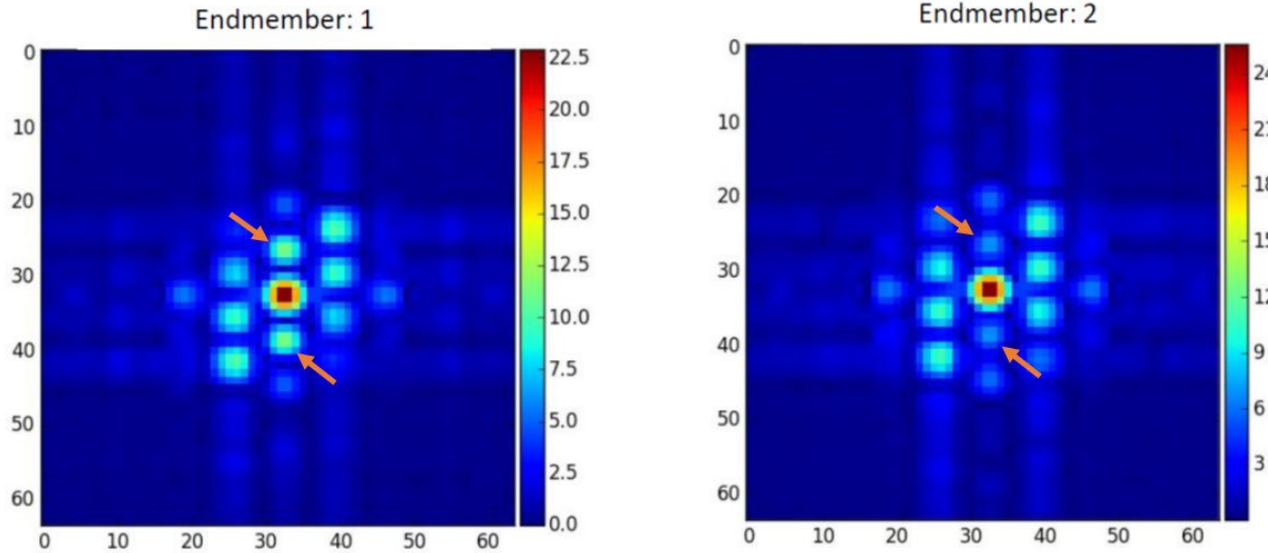
Real-space abundance maps



In a good agreement with test case, 2 spots in the “inner hexagon” are strongly suppressed in the 2nd component **reflecting a fine structure of charge ordered pattern**

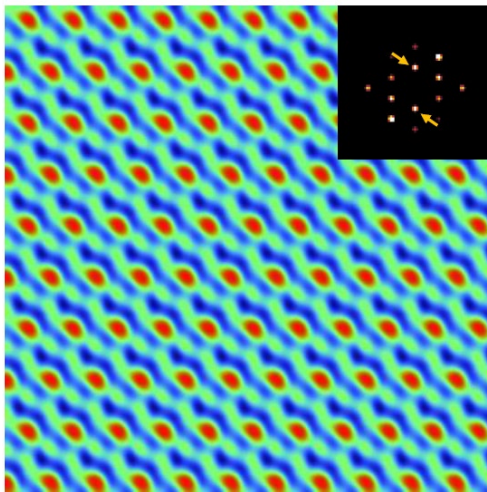
NFIND-R for coexisting order parameters

FFT endmembers

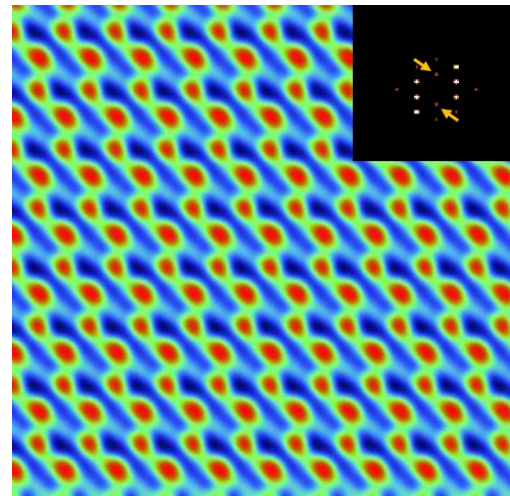


In the 2nd component, 2 spots in the “inner hexagon” are strongly suppressed reflecting a fine structure of charge ordered pattern

Real-space images of corresponding phases



Hexagonal superlattice



Dimer superlattice

M. ZIATDINOV, A. BANERJEE, A. MAKSOV, T. BERLIJN, W. ZHOU, H.B. CAO, J.Q. YAN, C.A. BRIDGES, D.G. MANDRUS, S.E. NAGLER, A.P. BADDORF, and S.V. KALININ, *Atomic-scale observation of structural and electronic orders in the layered compound α -RuCl₃*, Nature Comm. 7, 13774 (2016).

- We always have a problem of window size:
 - too large – loose spatial resolution,
 - too small – FFT behaves poorly due to edge effects
- Interpretation of FFT data is complicated (too much data if fit each peak, unclear meaning of the unmixing components)
- Natural descriptor for atomically resolved images – atomic coordinates!

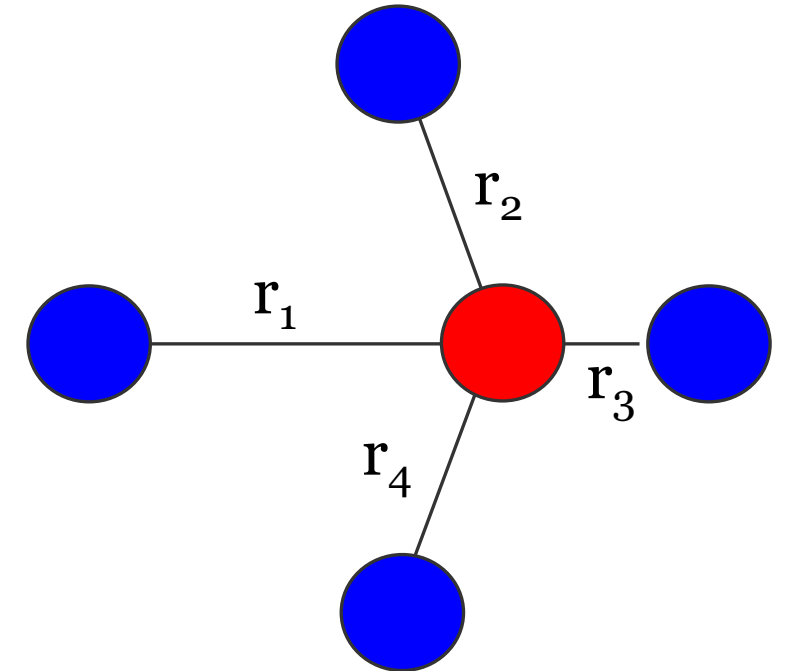
For each atom, define nearest neighbors and generate array of the corresponding radius-vectors of the form

$$NA_{ij} = (rx_1, ry_1, rx_2, ry_2, rx_3, ry_3, rx_4, ry_4)_{ij}$$

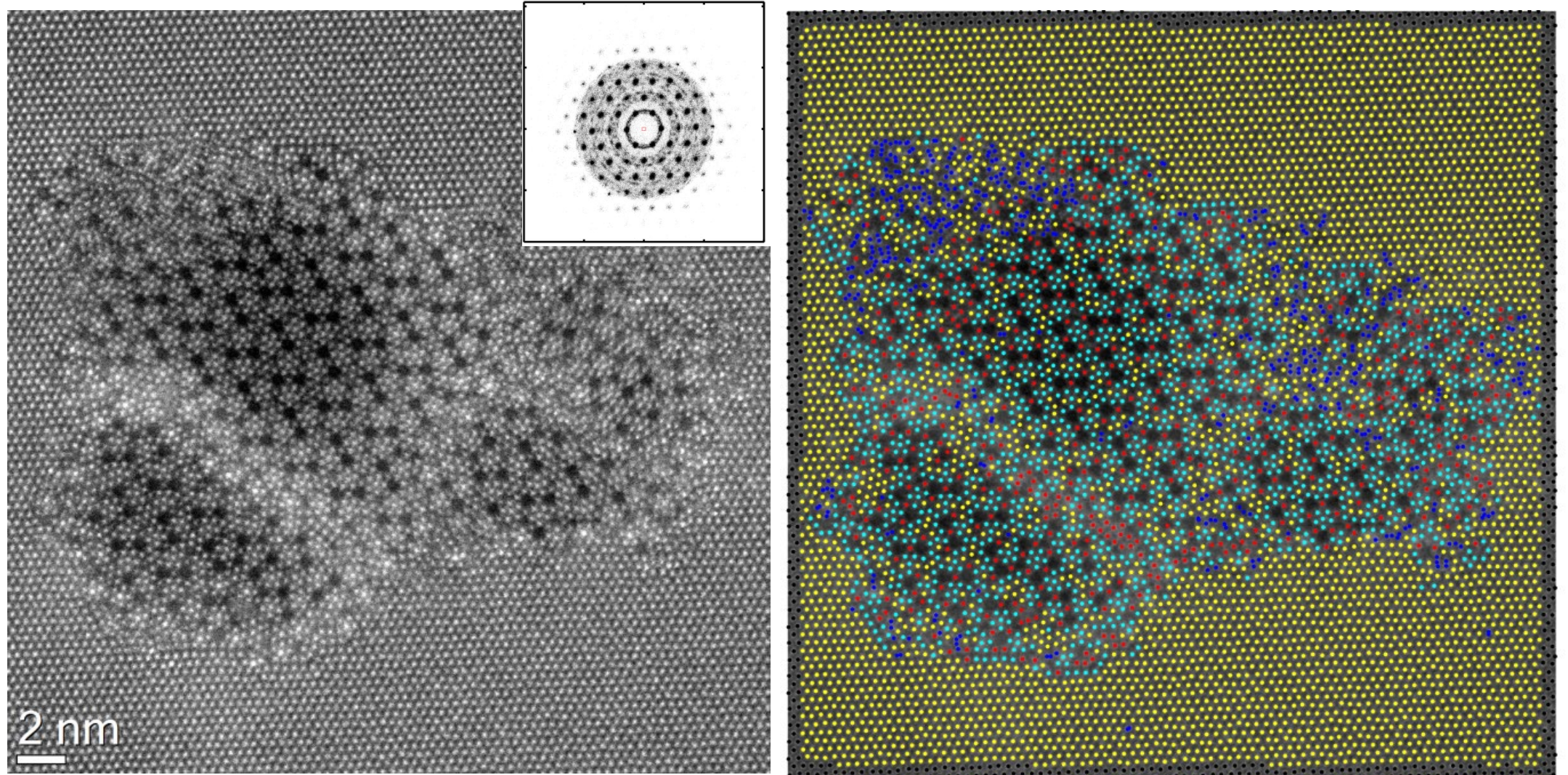
Indexes 1,2,3,4 are chosen in the same sense for all atoms (generalization for different lattice and/or next coordination sphere obvious)

Then, phase/ferroic variant identification problem can be reduced to finding equivalent (in statistical sense) groups of nearest neighbors

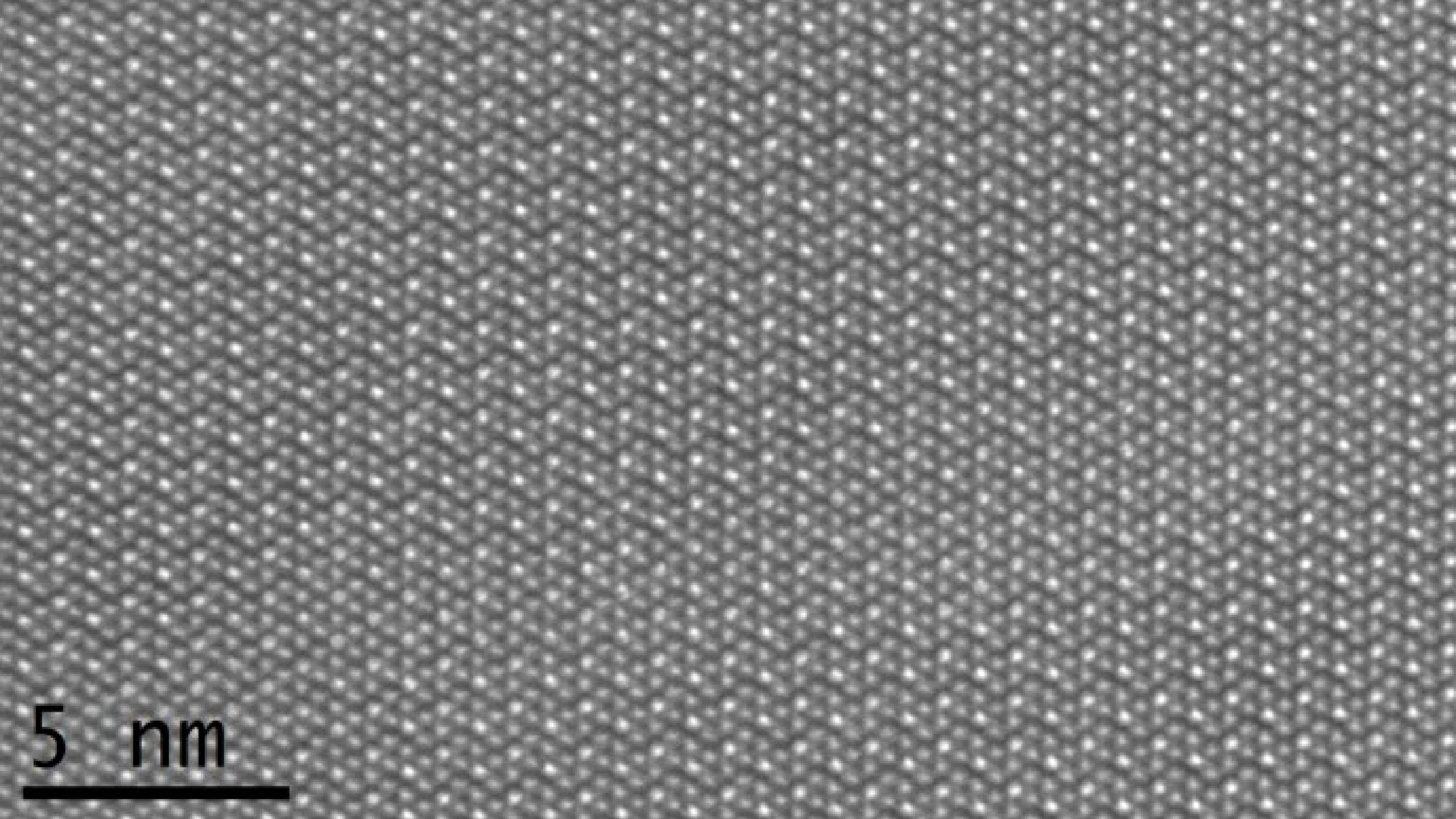
We can also use group theory to make hypotheses, e.g. add translation symmetry operations, i.e. $i \rightarrow i+1$ and $j \rightarrow j+1$ for lattice doubling)



Local crystallography: k-means



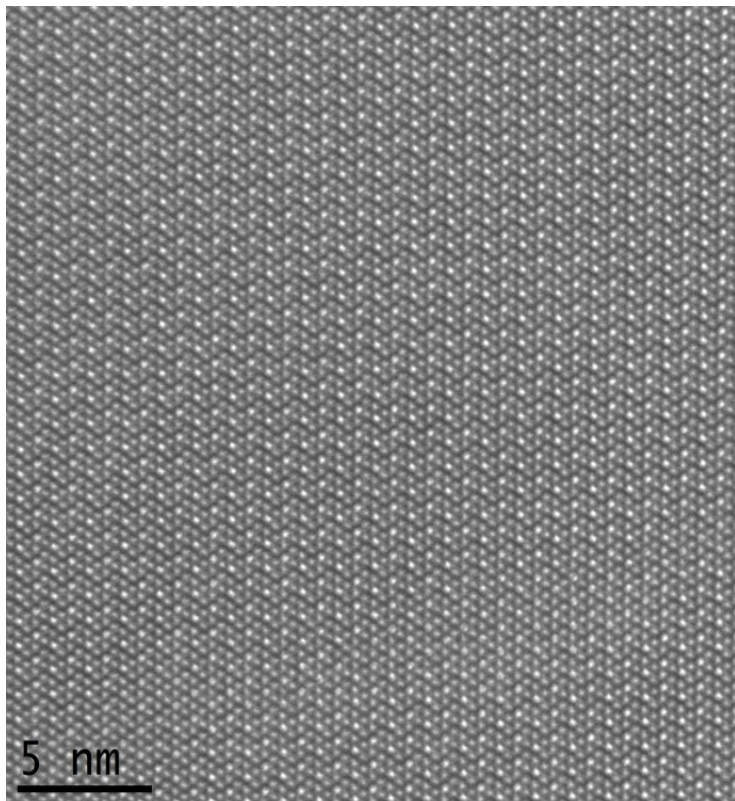
A. BELIANINOV, Q. HE, M. KRAVCHENKO, S. JESSE, A. BORISEVICH, and S.V. KALININ, *Identification of phases, symmetries, and defects through local crystallography*, Nat. Comm. **6**, 7801 (2015).



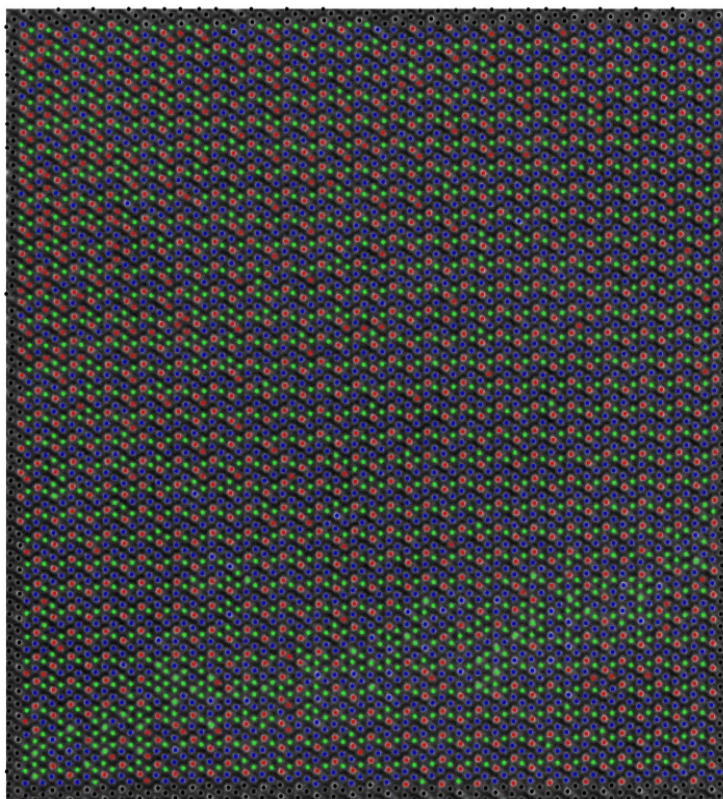
5 nm

Local crystallography

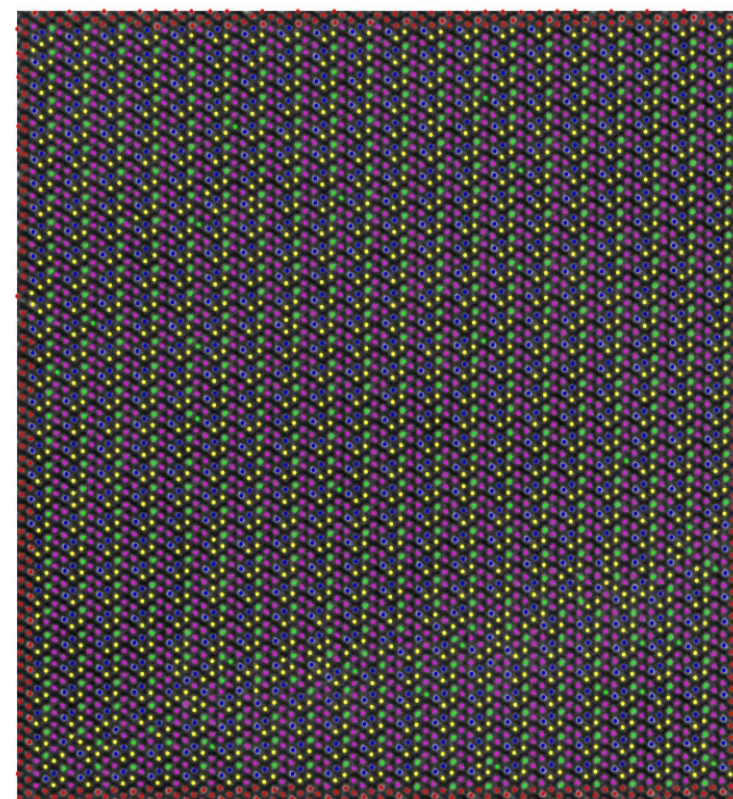
Image



K-means full vector



K-means angles



Normal modes: group theory

- **Group Representations** map group elements onto matrices, ensuring matrix multiplication aligns with the group operation.
- Molecules have **symmetries** defined by point groups linked to symmetry operations, such as rotations.
- Molecules have various **vibrational modes** with specific symmetries. Using the molecule's point group, one can deduce which modes are spectroscopically active.
- **Selection Rules:** Group representations dictate which vibrational modes appear in techniques like IR or Raman. Some modes may be IR-active but not Raman-active based on symmetry.
- **Vibrational frequencies** indicate energy differences between vibrational levels that are connected to group representations

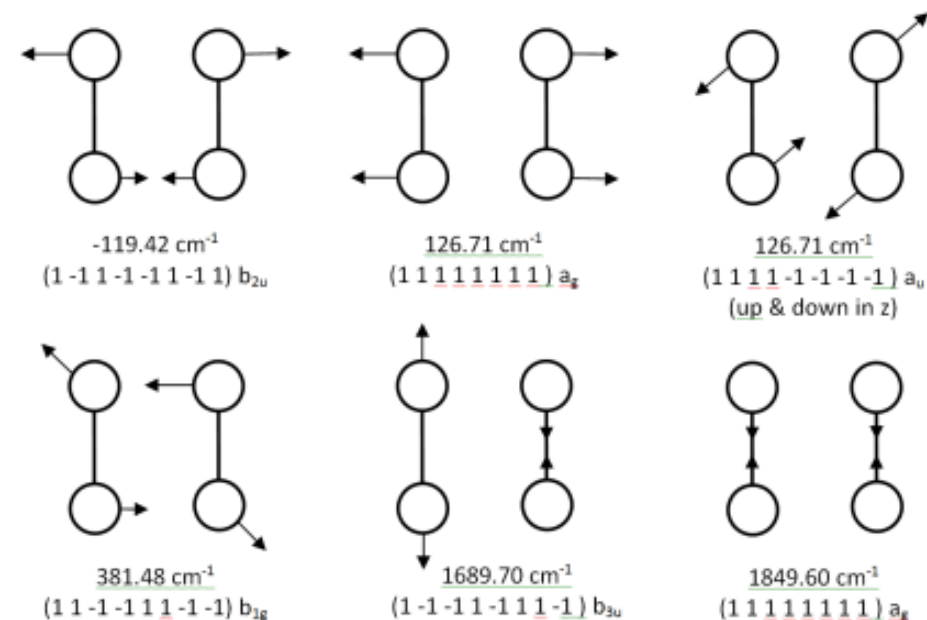
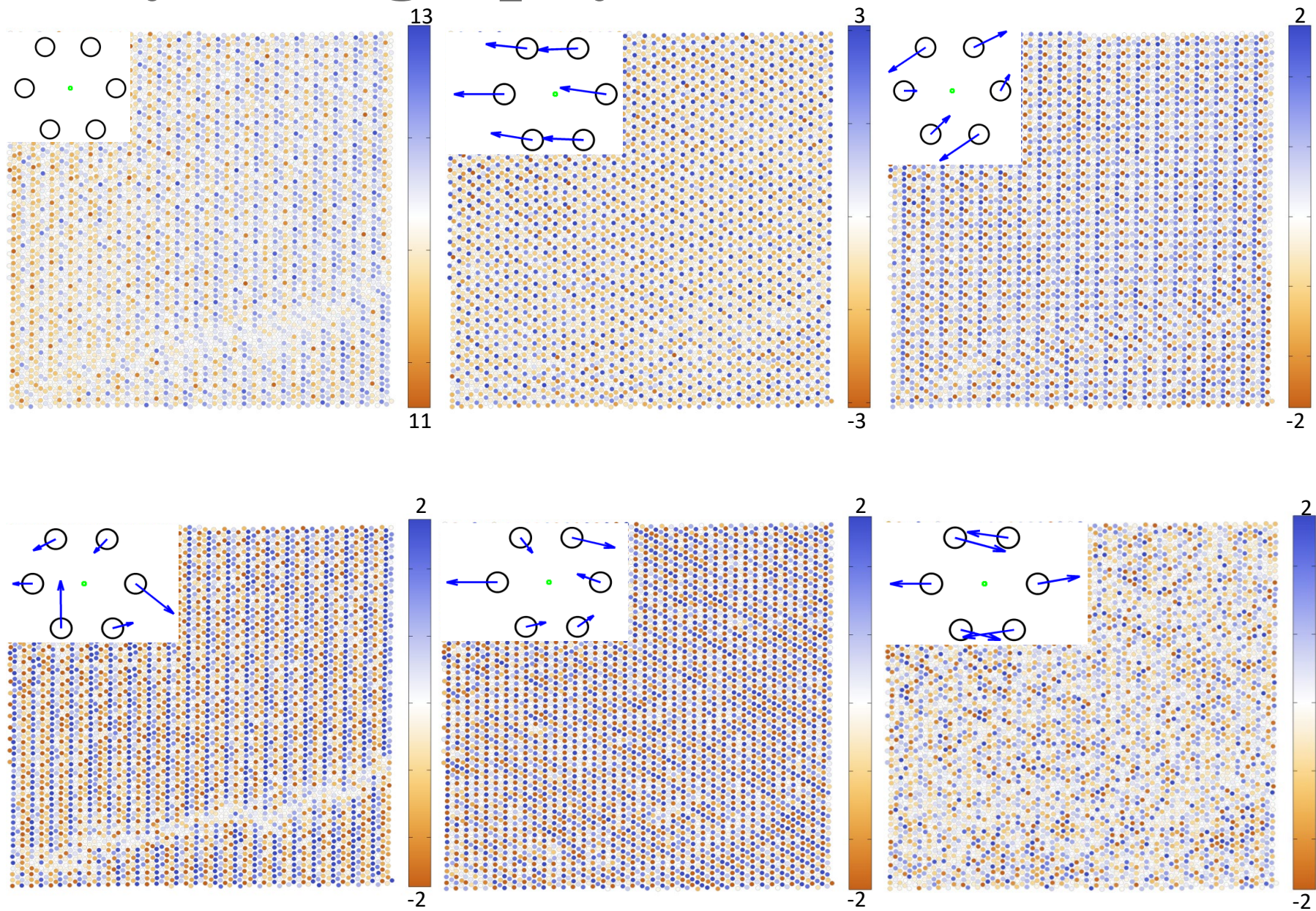


Figure 2: Sketches of Normal Modes of O_4^+

Table 2: Character Table for Point Group D_{2h}

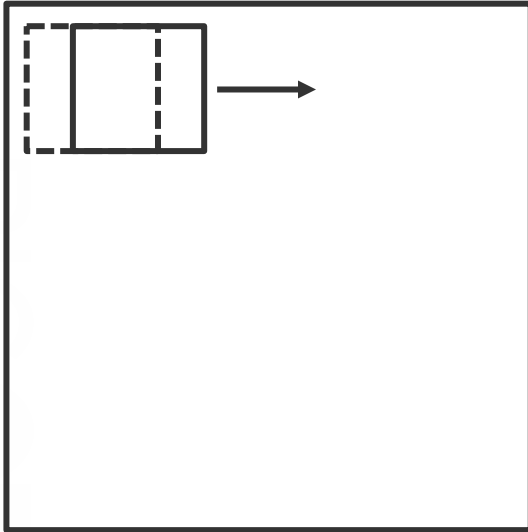
D_{2h}	E	$C_2(z)$	$C_2(y)$	$C_2(x)$	i	$\sigma(xy)$	$\sigma(xz)$	$\sigma(yz)$		
A_g	1	1	1	1	1	1	1	1	R_z R_y R_x	x^2, y^2, z^2 xy xz yz
B_{1g}	1	1	-1	-1	1	1	-1	-1		
B_{2g}	1	-1	1	-1	1	-1	1	-1		
B_{3g}	1	-1	-1	1	1	-1	-1	1		
A_u	1	1	1	1	-1	-1	-1	-1	z y x	
B_{1u}	1	1	-1	-1	-1	-1	1	1		
B_{2u}	1	-1	1	-1	-1	1	-1	1		
B_{3u}	1	-1	-1	1	-1	1	1	-1		

Local crystallography: PCA

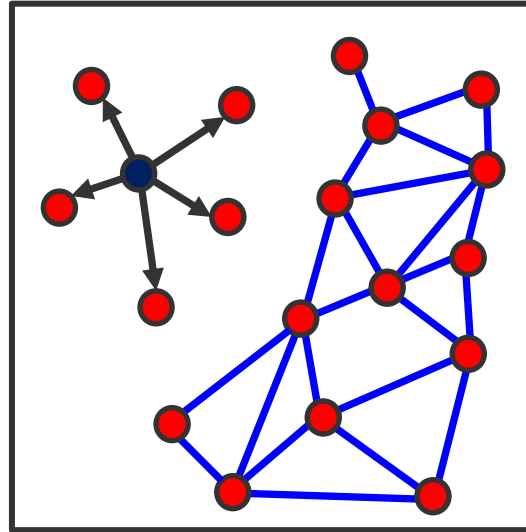


Constructing the descriptors

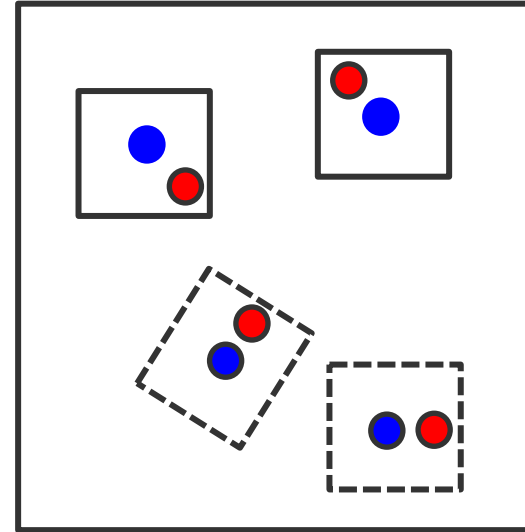
**Continuous
translational
symmetry**



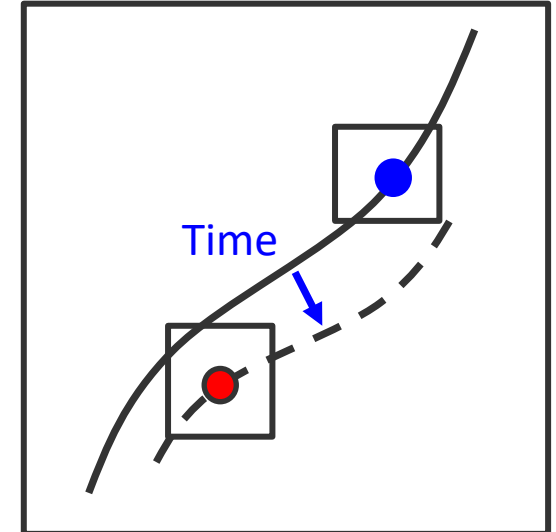
**Atom based
descriptions**



**Localized
sub-images**



**Time-delayed
descriptors**

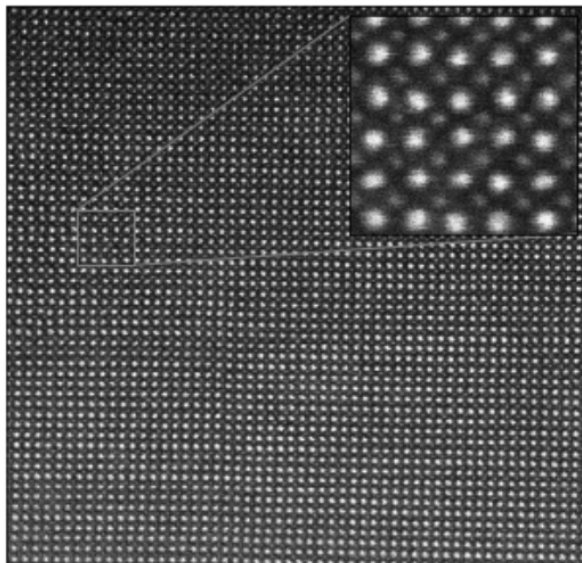


The choice of the descriptor:

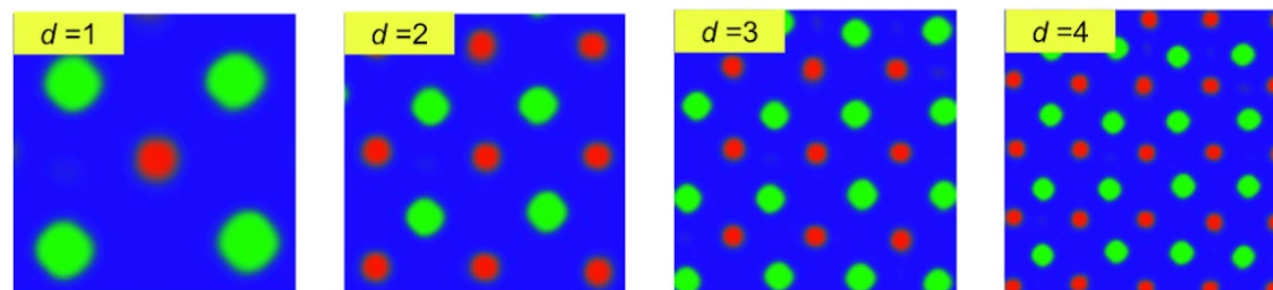
- Defines physical inferential biases and allows to introduce prior knowledge
- Determines the physical meaning of the analysis
- Establishes the analysis pipeline

Local crystallography: FerroNET

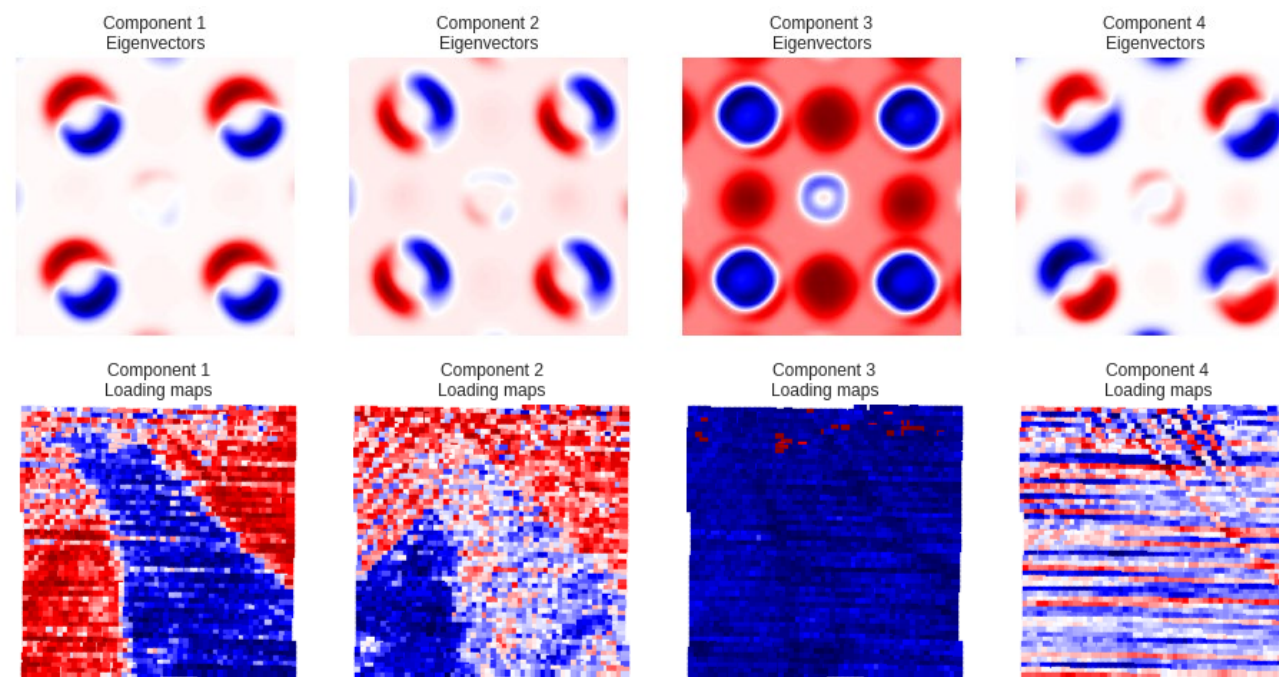
Experimental (LBFO film)



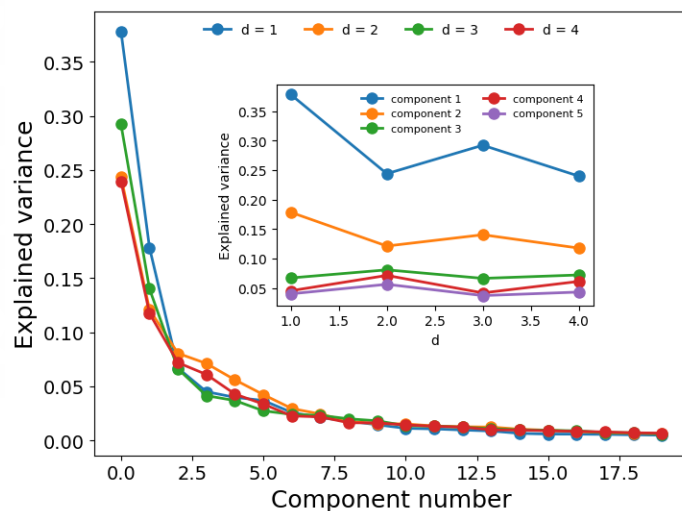
Building blocks (from neural network output)



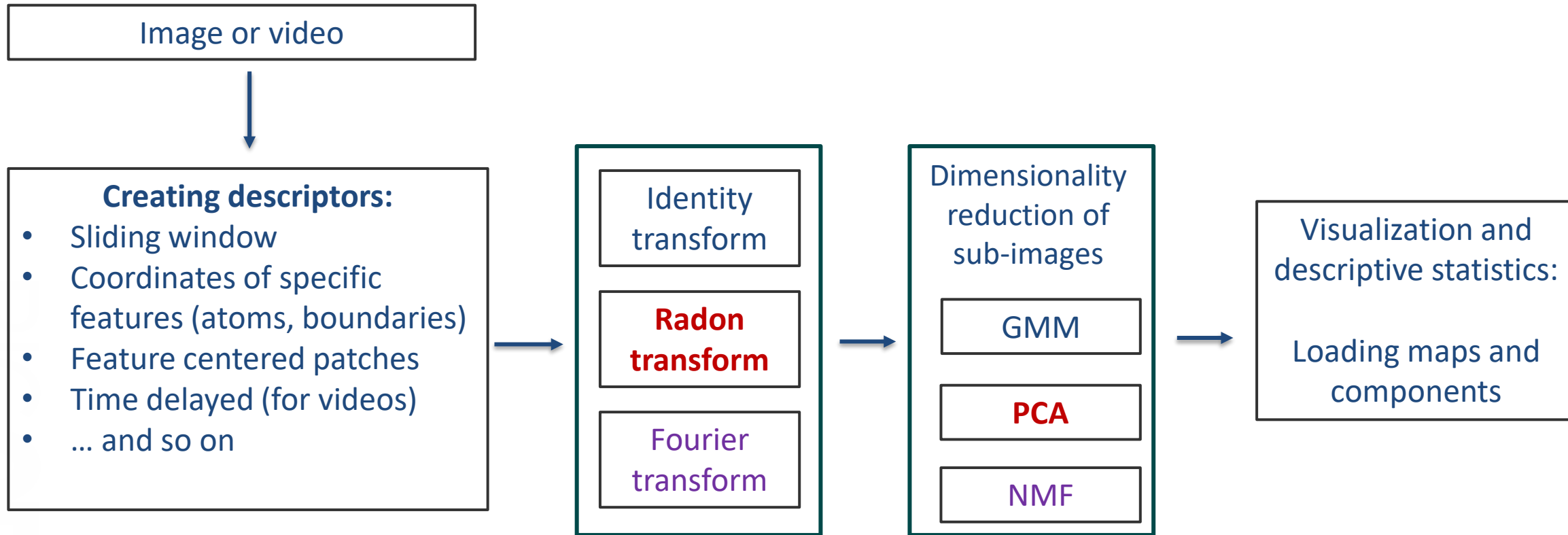
PCA eigenvectors and loading maps



Information Content



Example of analysis pipeline



Pipelines are defined to

- Make analysis traceable, repeatable, explainable, and transferable
- Allow for hyperparameter tuning and optimization
- Efficiently use the memory

How general should you be: depends on applications

And one transform to rule them all...

